Sarvan Sir's Economy Notes

NATIONAL INCOME ACCOUNTING

GDP – Product Method (Value Added Method / Output Method)

Gross Domestic Product (GDP)

is the total value of all final goods and services produced within the domestic territory of a country by resident units during a specific period of time.

1. Resident Unit

An individual or institutional unit having a predominant economic interest within the economic territory of a country and generally performing economic activities for more than one year.

Examples:

1. A foreign employee staying and working in India for two years is generally considered a resident unit of India.
2. A foreign tourist visiting India for two months is considered a non-resident unit.
3. An Indian citizen permanently settled in the USA is considered a non-resident unit of India.

2. Domestic Economic Territory

Includes:

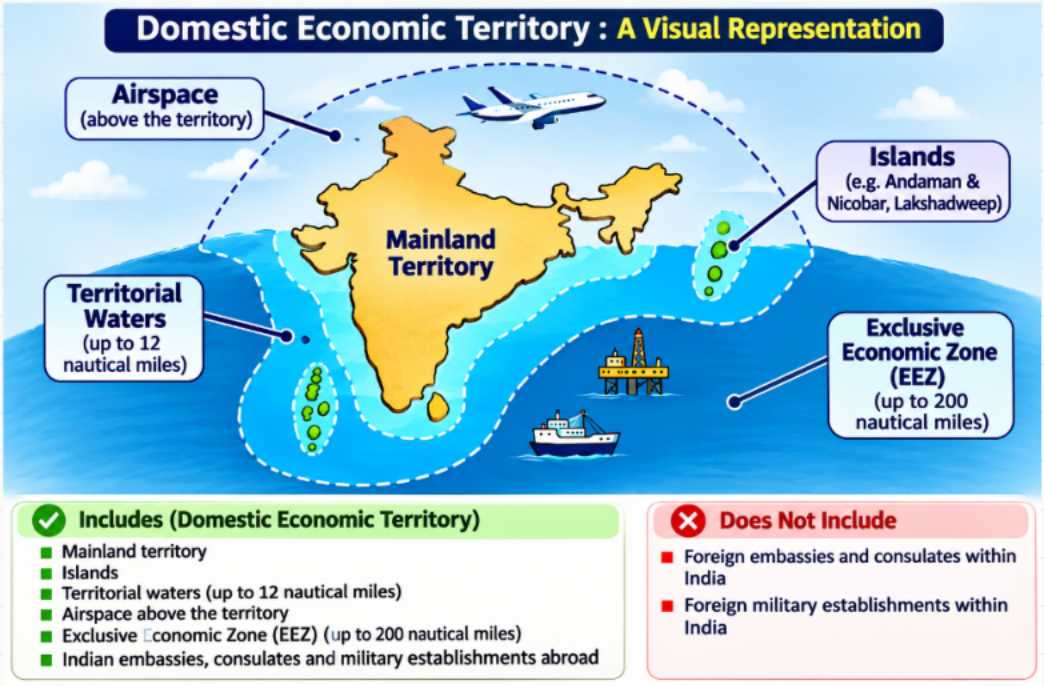
- Mainland territory of the country.
- Islands belonging to the country.
- Territorial waters.
- Airspace above the domestic territory.
- Exclusive Economic Zone (EEZ), to the extent that the country exercises exclusive economic rights over resources and activities.
- Indian embassies, consulates and military establishments located in foreign countries.

Does not include:

- Foreign embassies and consulates located within India.
- Foreign military establishments located within India.

Note: Economic territory is not exactly the same as geographical territory. Indian embassies abroad are treated as part of India's economic territory, whereas foreign embassies in India are treated as part of the respective foreign country's economic territory.

Domestic Economic Territory – Visual Representation



Sarvan Sir's Economy Notes

Product Method / Value Added Method / Output Method

1. Meaning / Definition

- Under this method, GDP is calculated by aggregating the value added at each stage of production (or the value of final goods and services) to arrive at the total value of output produced in the economy.

2. Formula (GDP at Market Price)

GDP (at Market Price) = GVA (at Basic Prices) + Net Product Taxes

3. GVA at Basic Prices – Formula (MoSPI)

GVA (at Basic Prices) = CE + OS/MI + CFC + Taxes on Production – Subsidies on Production

4. Components of GVA (Factor Payments)

CE	**Compensation to Employees** – Salaries, wages and employers' social contributions.
OS	**Operating Surplus** – Income of incorporated enterprises from production (profit, rent and interest).
MI	**Mixed Income** – Income of self-employed persons and owners of unincorporated enterprises (combining labour and capital).
CFC	**Consumption of Fixed Capital** – Depreciation on fixed assets due to wear and tear, ageing and obsolescence.
Taxes on Production	Taxes levied on production by the government, such as property tax, business licence fees, payroll tax, etc.
Subsidies on Production	Financial assistance provided by the government to producers towards production-related costs and employment.

5. Meaning of GVA at Basic Prices

- **Gross Value Added (GVA) at Basic Prices** is the value of output produced by an economic unit, measured at basic prices (producer's price), minus the value of intermediate consumption.

6. Alternative Formula

GVA at Basic Prices = Output – Intermediate Consumption

Basic Price (Producer's Price):

- The amount receivable by the producer from the purchaser for a unit of output, excluding taxes on products and including subsidies on products.

Important Note – Basic Price vs Market Price

- **Basic Price:** Excludes taxes on products (e.g., GST, excise duty and customs duty) and includes subsidies on products.
- **Market Price:** Includes taxes on products and excludes subsidies on products.

Sarvan Sir's Economy Notes

NATIONAL INCOME ACCOUNTING

Illustration : GDP Calculation – Farmer and Bakery (Product Method)

Example : Farmer – Bakery – Consumer

Let us understand the calculation of GDP and product tax with a simple example of a farmer and a bakery.

Step 1 : Farmer – Production and Sale of Wheat

- Total wheat produced = ₹1,000
- Retained for own consumption = **₹600**
- Sold to bakery owner = ₹400
- GST @ 10% on **₹400** = ₹40
- Farmer receives **₹440** (₹400 + ₹40)
- Farmer remits GST of **₹40** to Government.

Step 2 : Bakery – Value Addition and Output

- Input (Wheat) purchased = ₹440 (₹400 + ₹40 GST @ 10%)
- Value added by the bakery = **₹500**
- Value of bread (before GST) = ₹400 + ₹500 = **₹900**
- Add GST @ 10% on ₹900 = ₹90
- Final price to consumer = ₹990
- The ₹40 (GST paid on input) is an Input Tax Credit (ITC) and not included again while computing output tax.

Value of Bread (before GST) = ₹900

↓

GST @ 10% = ₹90

↓

Final Price = ₹990

Step 3 : GST Settlement, Government Receipt and GDP Calculation

(A) GST Settlement by Bakery

Output tax collected (on bread)	₹90
Input tax already paid (on wheat)	₹40
Net GST payable to Government (₹90 – ₹40)	**₹50**

(B) Total Product Tax Received by Government

From Farmer	₹40
From Bakery (Net) =	₹50
Total Tax Received =	₹90

Government receives the tax only on the final goods = ₹90.

(C) Calculation of GDP

- **GDP :** The final value of all goods and services (monetary value) produced in the economy.

Final Goods	Value (₹)
Wheat (own consumption)	600
Bread (sold to ultimate consumer)	990
Total GDP	**1,590**

Conclusion : GDP of this economy = **₹1,590** and Product tax received by the Government = **₹90.**

Key Takeaway – Product Method

The value added at different stages of production, when aggregated, gives the total value of final goods and services produced in the economy. This method avoids double counting of intermediate goods.

Sarvan Sir’s Economy Notes

NATIONAL INCOME ACCOUNTING

PRODUCT METHOD – CONCLUDING POINTS

1. DIFFERENCE BETWEEN GVA AND GDP

GVA (Gross Value Added)

GVA measures the value added by a producer or production unit during the process of production. It is calculated as the value of output minus intermediate consumption.

GDP (Gross Domestic Product)

GDP measures the total monetary value of all final goods and services produced within the domestic territory of a country during a specific period.

2. CHANGE IN INDIA’S NATIONAL ACCOUNTS – 2015

Adoption of the 2008 SNA

In 2015, India introduced major changes to its National Accounts compilation methodology and aligned it with the United Nations’ System of National Accounts 2008 (2008 SNA).

SNA = System of National Accounts

Sarvan Sir's Economy Notes

GDP Calculation – Product Method

PRELIMS PRACTICE SET – 1 | *15 MCQs* | *APPSC Group 1 / UPSC Level*

Instructions: Choose the most appropriate answer. The answer key and brief explanations are provided after all 15 questions.

Section A – Conceptual Understanding

1. GDP under the Product Method is calculated by:

A. Adding all factor incomes generated in an economy.

B. Adding expenditure incurred on final goods and services.

C. Aggregating the value added by all producing units in an economy.

D. Adding all monetary transactions undertaken during an accounting year.

2. Consider the following statements regarding Value Added:

1. Value Added is the difference between the value of output and intermediate consumption.
2. Value Added includes the value of all intermediate goods purchased by a producing unit.
3. Aggregating value added across producing units helps avoid double counting.

A. 1 only B. 1 and 3 only C. 2 and 3 only D. 1, 2 and 3

3. Consider the following activities:

1. Purchase of wheat by a flour mill.
2. Purchase of flour by a bakery.
3. Purchase of bread by a household.
4. Purchase of machinery by a manufacturing enterprise.

Which of the above are generally treated as final goods?

A. 1 and 2 only B. 3 and 4 only C. 1, 3 and 4 only D. 2, 3 and 4 only

4. The primary objective of using the Value Added Method in GDP calculation is to:

A. Eliminate the impact of inflation.

B. Avoid double counting of intermediate goods.

C. Exclude all capital goods from GDP.

D. Eliminate the contribution of the service sector.

Section B – Application and Statement-Based Questions

5. Consider the following statements regarding GDP calculation in India's national accounting framework:

1. GVA at Basic Prices is obtained by subtracting intermediate consumption from the value of output.
2. Net Taxes on Products are added to GVA at Basic Prices to arrive at GDP at Market Prices.
3. Net Taxes on Production and Net Taxes on Products are identical concepts.

A. 1 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

6. Consider the following production chain:

Farmer (wheat): output ■100; Flour mill (flour): output ■160; Bakery (bread): output ■250.

Assuming no other intermediate consumption, what is the total value added in this chain?

A. ■510 B. ■410 C. ■250 D. ■160

7. Consider the following statements regarding unsold goods:

1. Goods produced during an accounting year but remaining unsold are excluded from GDP.
2. Changes in inventories are considered in national income accounting.
3. Unsold finished goods are treated as additions to inventories.

A. 1 only B. 2 only C. 2 and 3 only D. 1, 2 and 3

8. Which of the following is correctly treated while calculating GDP?

A. The entire value of a second-hand car sold between two individuals is added to current GDP.

B. The value of a second-hand car is included in GDP again when ownership changes.

C. The brokerage or commission earned by a dealer on the sale of a second-hand car is included in current GDP.

D. All transactions involving second-hand goods are excluded from GDP, including services provided by dealers.

9. Which of the following is NOT included in GDP under the Product Method?

A. Value of newly produced machinery.

B. Value of government-provided non-market services.

C. Value of newly produced consumer goods.

D. Pension payments made by the government to retired individuals.

10. Consider the following statements:

1. Capital goods produced by an enterprise for its own use can be included in GDP.
2. Unpaid household services produced and consumed within the same household are generally excluded from GDP.
3. Imputed rental value of owner-occupied houses is included in national income accounting.

A. 1 only B. 1 and 2 only C. 1 and 3 only D. 1, 2 and 3

Section C – Advanced Prelims Questions

11. Government-provided non-market services such as public administration and defence are generally valued in national income accounting on the basis of:

A. Their market selling price.

B. The cost of production.

C. Their contribution to exports.

D. The tax revenue collected by the government.

12. Consider the following statements regarding imported intermediate goods:

1. Imported intermediate goods used in domestic production are included in intermediate consumption.
2. Their value is deducted while calculating domestic value added.
3. Importing intermediate goods automatically increases domestic GDP by the value of imports.

A. 1 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

13. Consider the following statements regarding taxes in GDP calculation:

1. Net Taxes on Products are added to GVA at Basic Prices to arrive at GDP at Market Prices.
2. Taxes on Products include taxes such as GST and excise duties.
3. Subsidies on Products are added while calculating Net Taxes on Products.

A. 1 only B. 1 and 2 only C. 2 and 3 only D. 1, 2 and 3

14. Consider the following statements:

1. Gross Value Added includes Consumption of Fixed Capital.
2. Net Value Added is obtained by deducting Consumption of Fixed Capital from Gross Value Added.
3. Depreciation is deducted while calculating Gross Value Added.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

15. Which of the following transactions would NOT directly contribute to the current year's GDP?

A. Production of new machinery.

B. Production of agricultural commodities.

C. Purchase of existing shares in a stock exchange.

D. Production of newly constructed residential buildings.

Answer Key and Explanations

Attempt all questions before referring to this section.

1 **C**	**2** **B**	**3** **B**	**4** **B**	**5** **B**
6 **C**	**7** **C**	**8** **C**	**9** **D**	**10** **D**
11 **B**	**12** **B**	**13** **B**	**14** **A**	**15** **C**

Q1. Answer: C

The Product Method aggregates the value added generated by producing units across the economy.

Q2. Answer: B

Value Added = Value of Output – Intermediate Consumption. Statement 2 is incorrect because intermediate consumption is deducted, not added.

Q3. Answer: B

Wheat and flour are intermediate inputs in the stated production chain. Bread bought by a household and machinery acquired as a capital good are final goods.

Q4. Answer: B

The method counts the value added at each stage rather than repeatedly counting the full value of intermediate transactions.

Q5. Answer: B

Statements 1 and 2 are correct. Net Taxes on Production and Net Taxes on Products are distinct components.

Q6. Answer: C

Total value added = ■100 + (■160 – ■100) + (■250 – ■160) = ■250. It equals the value of the final output in this simplified chain.

Q7. Answer: C

Unsold current production is counted as inventory investment. Therefore, statements 2 and 3 are correct and statement 1 is incorrect.

Q8. Answer: C

The used car itself was counted when originally produced. A current dealer's commission is a newly provided service and is included.

Q9. Answer: D

Pensions are transfer payments. They are not payments for current production of goods or services.

Q10. Answer: D

All three statements are correct. Own-account production of capital goods is included in GDP. Unpaid domestic services performed within households are generally excluded, while imputed rental services of owner-occupied dwellings are included in national income accounting.

Q11. Answer: B

Since non-market services do not have an observable market price, their output is generally valued using the cost of production.

Q12. Answer: B

Imported inputs form part of intermediate consumption and are deducted in calculating domestic value added. The import itself is not domestic production.

Q13. Answer: B

Net Taxes on Products = Taxes on Products – Subsidies on Products. Therefore, statement 3 is incorrect.

Q14. Answer: A

GVA is a gross measure and includes Consumption of Fixed Capital (CFC). NVA = GVA – CFC.

Q15. Answer: C

Purchase of existing shares is a transfer of ownership of a financial asset, not current production. Brokerage services, if any, are counted separately.

Quick Formula Revision
GVA = Value of Output – Intermediate Consumption
NVA = GVA – Consumption of Fixed Capital
GDP at Market Prices = Σ GVA at Basic Prices + Net Taxes on Products
Net Taxes on Products = Taxes on Products – Subsidies on Products

Sarvan Sir's Economy Notes

Investment

1 Meaning / Concept

Investment means the total value of the capital goods that form part of an economy's final output (GDP).

In simple terms: Investment represents the value of capital goods.

2 Basic Formula

Investment = Value of Capital Goods

Gross Investment = Value of Capital Goods

For this simplified example, gross investment is measured by the value of capital goods included in the final output and acquired for productive use.

3 Numerical Illustration (Rs.)

Suppose a factory is established with an initial value of Rs.1,00,000. During the accounting year, it produces final goods worth Rs.1,000.

Consumption Goods

Rs.700 = 70% of GDP

Capital Goods

Rs.300 = 30% of GDP

Total final output (GDP) = Rs.700 + Rs.300 = Rs.1,000

Gross Investment = Rs.300

Therefore, gross investment is 30% of GDP, while consumption goods account for 70% of GDP.

Investment – Imports, Exports & Capital Stock

4 Effect of Imports and Exports

A. Import of Capital Goods

Base domestic capital goods = Rs.300
Imported capital goods = Rs.100

Rs.300 + Rs.100 = Rs.400

Gross investment = Rs.400 (40% of GDP)

B. Export of Capital Goods

Base capital goods = Rs.300
Capital goods exported = Rs.50

Rs.300 − Rs.50 = Rs.250

Domestic investment = Rs.250 (25% of GDP)

Note: Imported capital goods add to domestic gross capital formation when acquired by resident producers. Capital goods exported are not retained as domestic investment. GDP is kept constant at Rs.1,000 in these simplified illustrations.

5 Gross Investment, Depreciation & Net Investment

Suppose gross investment is Rs.300 and depreciation (consumption of fixed capital) is Rs.50.

Net Investment = Gross Investment − Depreciation

Rs.300 − Rs.50 = Rs.250

6 Capital Stock

Opening capital stock (factory) = Rs.1,00,000
Add: Gross investment = Rs.300
Less: Depreciation = Rs.50

Closing Capital Stock = Opening Stock + Net Investment

Rs.1,00,000 + Rs.250 = Rs.1,00,250

NOTE

- Purchasing newly produced machinery is considered investment in macroeconomics because it adds to productive capital.
- Purchasing shares worth Rs.10 lakh is not investment in GDP accounting; it transfers ownership of a financial asset and does not itself create new capital goods.

SARVAN SIR'S ECONOMY NOTES

GDP CALCULATION

Income Method & Expenditure Method | 15 Prelims MCQs

Level: UPSC / APPSC Group 1	**Coverage:** Income, expenditure, commodity-flow, fund-flow, inventories and valuables

Instructions

Choose the most appropriate answer for each question. The answer key and explanations are provided after all 15 questions.

Section A – Income Method

1. Consider the following components of income under the income method of calculating GDP:

1. Compensation of employees
2. Operating surplus
3. Mixed income
4. Transfer payments

A. 1 and 2 only
B. 1, 2 and 3 only
C. 1, 3 and 4 only
D. 1, 2, 3 and 4

2. Consider the following statements regarding mixed income:

1. It is generally associated with self-employed persons and unincorporated enterprises.
2. It includes both the labour income and entrepreneurial income of the proprietor when these cannot be separately identified.
3. It represents only the interest earned by the owner of an enterprise.

A. 1 only
B. 1 and 2 only
C. 2 and 3 only
D. 1, 2 and 3

3. Consider the following statements regarding consumption of fixed capital (CFC):

1. CFC represents the decline in the value of fixed assets due to normal wear and tear and obsolescence.
2. CFC is deducted from Gross Value Added to obtain Net Value Added.
3. CFC is treated as a transfer payment.

A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

4. An economy records the following factor incomes during a particular year: Compensation of employees: ■500 crore; Operating surplus: ■200 crore; Mixed income: ■100 crore. Assuming these are the complete income components, what is the total factor income generated?

A. ■600 crore

B. ■700 crore

C. ■800 crore

D. ■900 crore

Section B – Expenditure Method

5. Which of the following correctly represents the expenditure method of calculating GDP at market prices?

A. PFCE + GFCE + GCF + Exports – Imports

B. PFCE + GFCE + GCF – Exports + Imports

C. Compensation of employees + Operating surplus + Mixed income

D. PFCE + Intermediate consumption + Exports – Imports

6. Consider the following items:

1. Private Final Consumption Expenditure
2. Government Final Consumption Expenditure
3. Gross Capital Formation
4. Net Exports

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Section C – Commodity-Flow and Fund-Flow Approaches

7. What is the fundamental principle of the commodity-flow approach used in estimating expenditure on commodities?

A. Tracking the movement of money between financial institutions

B. Tracing the supply of commodities and allocating them among their different uses

C. Calculating only the value of commodities exported

D. Adding the sales value at every stage of production

8. Consider the following statements regarding the commodity-flow approach:

1. It can be used to estimate expenditure on machinery and equipment.
2. It considers the availability of commodities from domestic production and imports.
3. It ignores the distinction between intermediate and final uses.
4. It helps estimate the allocation of commodities among different uses.

A. 1 and 2 only

B. 1, 2 and 4 only

C. 2 and 3 only

D. 1, 2, 3 and 4

9. Consider the following statements regarding fund-flow analysis:

1. It focuses on tracing financial resources and their uses.
2. It is conceptually different from tracing the physical flow of commodities.
3. It necessarily includes every financial transaction as part of GDP.

A. 1 only

B. 1 and 2 only

C. 2 and 3 only

D. 1, 2 and 3

Section D – Inventories, Valuables and Gross Capital Formation

10. Consider the following components of Gross Capital Formation (GCF):

1. Gross Fixed Capital Formation
2. Change in inventories
3. Net acquisition of valuables
4. Purchase of existing shares in the stock market

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

11. Consider the following statements regarding inventories:

1. Unsold goods produced during the current accounting period are included in inventory investment.
2. A decrease in inventories can be recorded as negative inventory investment.
3. Only finished goods are included in inventories; raw materials and work-in-progress are excluded.

A. 1 only

B. 1 and 2 only

C. 2 and 3 only

D. 1, 2 and 3

12. A firm produces goods worth ■500 crore during a particular year. It sells goods worth ■420 crore and adds the remaining ■80 crore to its inventories. Assuming no other inventory changes, what is the contribution of this production to GDP through sales and inventory accumulation?

A. ■420 crore

B. ■460 crore

C. ■500 crore

D. ■580 crore

13. Consider the following statements regarding valuables in national income accounting:

1. Gold, precious stones and jewellery may be treated as valuables.

2. Net acquisition of valuables is included in Gross Capital Formation.

3. Purchase of existing financial assets such as shares is included in net acquisition of valuables.

4. Valuables are generally held as stores of value.

A. 1 and 2 only

B. 1, 2 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

14. Consider the following transactions:

1. Purchase of a newly manufactured machine by a factory.

2. Purchase of an existing residential house by a household.

3. Purchase of newly produced gold jewellery as a valuable.

4. Purchase of existing shares from another investor.

A. 1 and 2 only

B. 1 and 3 only

C. 1, 2 and 3 only

D. 1, 3 and 4 only

15. Consider the following statements regarding the treatment of second-hand goods in GDP calculation:

1. The resale value of an existing second-hand car is included in the current year's GDP.

2. The commission earned by a dealer for facilitating the resale of a second-hand car is included in GDP.

3. The purchase of a newly manufactured car is included in GDP.

A. 1 only

B. 1 and 2 only

C. 2 and 3 only

D. 1, 2 and 3

ANSWER KEY

GDP Calculation – Income and Expenditure Methods

Q1: B	**Q2: B**	**Q3: A**	**Q4: C**	**Q5: A**
Q6: D	**Q7: B**	**Q8: B**	**Q9: B**	**Q10: B**
Q11: B	**Q12: C**	**Q13: B**	**Q14: B**	**Q15: C**

DETAILED EXPLANATIONS

Question 1 – Answer: B

Statements 1, 2 and 3 are factor-income components generated through current domestic production. Transfer payments, such as scholarships or pensions, are not payments for current productive services and are excluded.

Question 2 – Answer: B

Mixed income is generally associated with self-employed persons and unincorporated enterprises. It combines labour and entrepreneurial returns when they cannot be separately identified. Rent, interest and profit are components associated with operating surplus; mixed income is not only interest.

Question 3 – Answer: A

CFC represents the consumption or decline in value of fixed assets through normal wear and tear and obsolescence. It is deducted from GVA to obtain NVA. It is not a transfer payment.

Question 4 – Answer: C

Total factor income = Compensation of employees + Operating surplus + Mixed income = ■500 crore + ■200 crore + ■100 crore = ■800 crore.

Question 5 – Answer: A

At market prices, the expenditure identity is GDP = PFCE + GFCE + GCF + (Exports – Imports). Imports are deducted and exports are added.

Question 6 – Answer: D

All four are components of GDP under the expenditure approach: private final consumption, government final consumption, gross capital formation and net exports.

Question 7 – Answer: B

The commodity-flow approach traces the supply of a commodity and allocates it among its uses, such as intermediate consumption, final consumption, capital formation and exports. It does not simply add sales at every production stage.

Question 8 – Answer: B

The approach can trace domestic production and imports and allocate the available supply among uses, including machinery and equipment. Distinguishing intermediate from final uses is important; statement 3 is therefore incorrect.

Question 9 – Answer: B

Fund-flow analysis focuses on financial resources and their uses, whereas commodity-flow analysis follows goods. Not every financial transaction represents current production; for example, trading existing shares does not itself add to GDP.

Question 10 – Answer: B

GCF comprises GFCF, change in inventories and net acquisition of valuables. Purchases of existing shares are financial transactions and are not included in GCF.

Question 11 – Answer: B

Inventories include raw materials, work-in-progress and finished goods. Unsold current production adds to inventories; withdrawals or a decrease in inventories can produce negative inventory investment.

Question 12 – Answer: C

Sales of ■420 crore plus inventory accumulation of ■80 crore account for the full ■500 crore of current production. Unsold current output is recorded as inventory investment.

Question 13 – Answer: B

Gold, precious stones and jewellery may be treated as valuables when held as stores of value. Net acquisition of valuables is included in GCF. Existing shares are financial assets, not valuables for this purpose.

Question 14 – Answer: B

A newly manufactured machine is fixed capital formation, and newly produced jewellery acquired as a valuable can enter net acquisition of valuables. An existing house and existing shares are not current production or current GCF merely because ownership changes.

Question 15 – Answer: C

The resale value of an existing second-hand car is excluded to avoid counting its original production again. The dealer's current commission is a newly provided service and is included. A newly manufactured car is also included as current production.

Quick Formula Recap

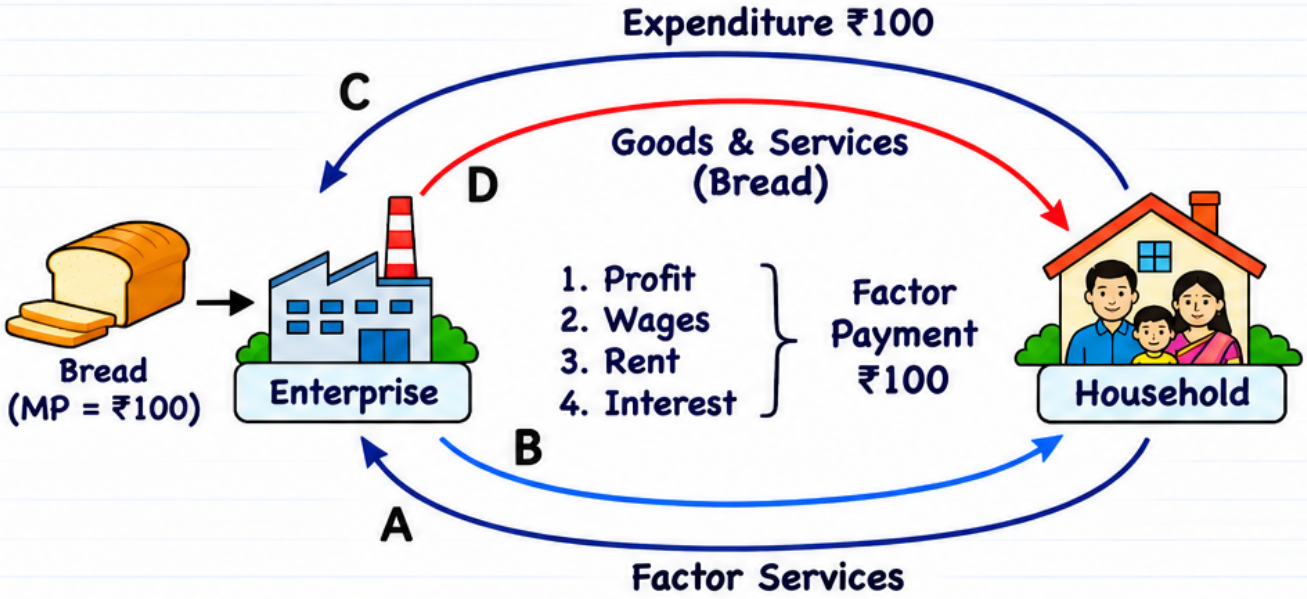
GDP at Market Prices = PFCE + GFCE + GCF + (Exports – Imports)
GCF = Gross Fixed Capital Formation + Change in Inventories + Net Acquisition of Valuables
NVA = GVA – Consumption of Fixed Capital (CFC)

Remember: Inventory investment is the change in stocks, not the total stock of inventories. Purchases of existing financial assets such as shares are not gross capital formation.

Sarvan Sir's Economy Notes

The Circular Flow of Income in an Economy

Case 1 : Two-Sector Economy



Explanation of the Flows

A : Households supply factor services (land, labour, capital and entrepreneurship) to enterprises.

B : Enterprises make factor payments (wages, rent, interest and profit) to households. (Total = ₹100)

C : Households spend their income on the purchase of goods and services (bread) produced by enterprises. (Expenditure = ₹100)

D : Enterprises supply goods and services (bread) to households.

Meaning / Concept

- The circular flow of income refers to the continuous movement of goods and services, factor services and money payments between different sectors of an economy.
- In a two-sector economy, it operates between two sectors only – **Households and Enterprises** (Firms).

Key Points (Case 1)

- The enterprise produces a consumption good (bread) with a market price of ₹100.
- Factor payments made by the enterprise to households (profit, wages, rent and interest) add up to ₹100.
- Households spend the entire ₹100 on the purchase of bread.
- There are no savings and no leakages in this simple circular flow.

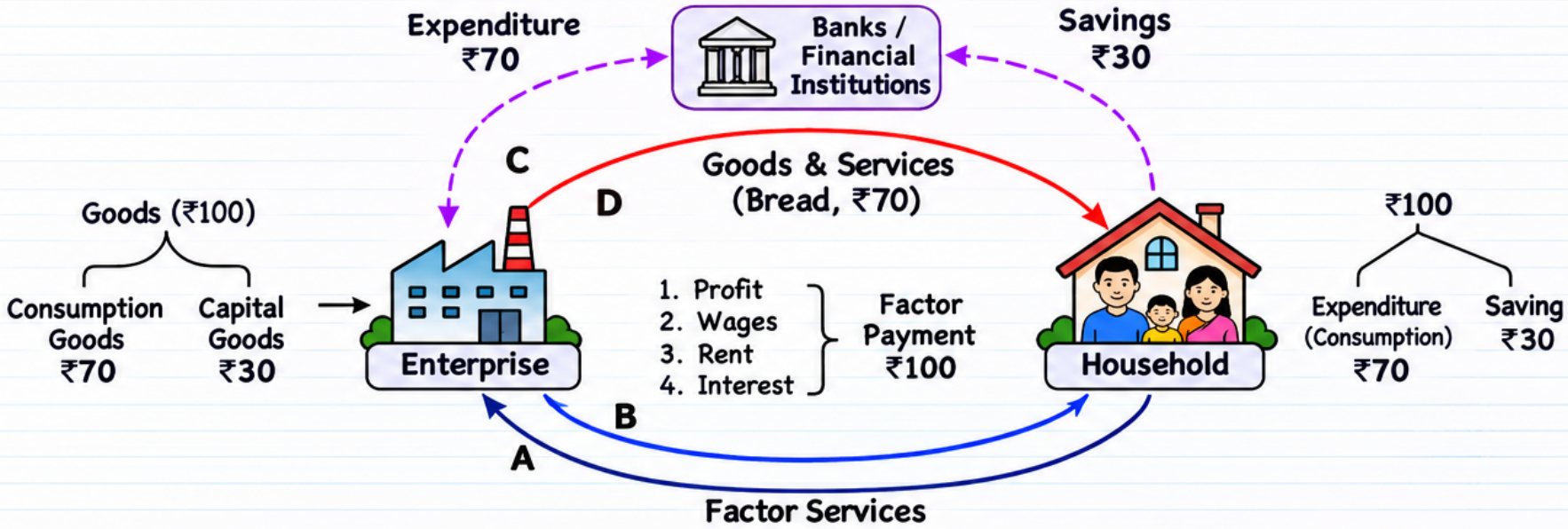
Conclusion

- The money (income) and the goods produced keep moving in a circle between households and enterprises.
- The total factor payments (income) are equal to the total expenditure on goods and services, i.e., ₹100.
- Hence, the value of output (₹100) is fully distributed as income in the economy.

Sarvan Sir's Economy Notes

The Circular Flow of Income in an Economy

Case 2 : Two-Sector Economy with Savings and Investment



1 Understanding the Flow

- In this case, households do not spend the entire income.
- Households spend ₹70 on consumption goods (bread) and save ₹30.
- The savings of households are deposited in banks/financial institutions.
- Banks, in turn, provide loans to enterprises to purchase capital goods worth ₹30.
- Thus, the economy produces both consumption goods (₹70) and capital goods (₹30).

2 Meaning of Arrows

A : Households supply factor services (land, labour, capital and entrepreneurship) to enterprises.

B : Enterprises make factor payments (wages, rent, interest and profit) of ₹100 to households.

C : Banks lend the savings (₹30) to enterprises for the purchase of capital goods (investment).

D : Enterprises supply goods and services (bread, worth ₹70) to households.

3 Key Points

- Total value of output = ₹100.
- Consumption goods = ₹70.
- Capital goods (Investment) = ₹30.
- Household savings (S) = ₹30.
- Investment (I) = ₹30.
- Hence, S = I.

4 Conclusion

- Savings of households are channelled through financial institutions (banks) into investment.
- This investment leads to the production of capital goods.
- Higher savings result in higher investment and a larger productive capacity, which promotes economic growth.

Sarvan Sir's Economy Notes

Savings, Investment and Growth : The Link (Circular Flow – Extension)

Key Idea

- Greater the savings in the economy, greater will be the production of capital goods.
- A part of the final output (capital goods) is called **investment.**
- Hence, savings (S) is equal to investment (I) in a closed, two-sector economy.
- Higher the savings, higher will be the investment and higher will be the production capacity of the economy, which will propel the economy on a **higher growth path.**

Flowchart : Savings to Higher GDP

More Savings in the Economy

↓

More **Loan Disbursal** by Financial Institutions (Banks)

↓

More Investment in the Economy

↓

More Production of **Capital Goods**

↓

In the Future, Higher Production of both **Capital Goods** and **Consumption Goods**

↓

Higher GDP (Economic Growth)

Explanation

- Households save a part of their income.
- These savings are deposited in financial institutions (banks), which in turn provide loans to enterprises.
- Enterprises use these loans for investment, i.e., for purchasing/producing capital goods (machines, tools, etc.).
- An increase in capital goods raises the productive capacity of the economy.
- With higher productive capacity, the economy can produce more goods and services in future, leading to a **higher GDP.**

Illustrative Examples

- In 1950, India was producing 95% consumption goods and only 5% capital goods (savings = 5%).
- In 2019-20, India's savings was around 28% of GDP, i.e., about 72% consumption goods and 28% capital goods.
- China has consistently been able to produce more than 40% capital goods out of its total output (GDP), which has propelled it into one of the fastest growing economies of the world.

Conclusion ➡ Higher savings → Higher investment → Higher production capacity → More output (GDP) → Higher economic growth.

Sarvan Sir's Economy Notes

National Income Accounting

Expenditure Method of Calculating GDP

1. What is GDP?

- **GDP (Gross Domestic Product)** is the total value of all **final goods and services** produced within the **domestic territory** of a country during a specific period.

2. Expenditure Method – Meaning

- Under this method, GDP is measured by adding the expenditure incurred by different sectors of the economy on final goods and services produced within the domestic territory during a given period.

3. Official Formula (as per MoSPI)

$$GDP\,(MP) = PFCE + GFCE + GCF + (X - M) + Discrepancies$$

Where:

- **PFCE** – Private Final Consumption Expenditure (Households)
- **GFCE** – Government Final Consumption Expenditure
- **GCF** – Gross Capital Formation (GFCF + Change in Inventories + Net Acquisition of Valuables)
- **X** – Exports of goods and services
- **M** – Imports of goods and services

Discrepancies:

- A statistical adjustment introduced by MoSPI.
- It captures the difference between GDP estimated by the product approach and the expenditure approach.
- It may be positive or negative.

Note: This is the official formula used by the Ministry of Statistics and Programme Implementation (MoSPI) for the estimation of GDP (at current and constant prices).

Gross Capital Formation (GCF)

- **Gross Capital Formation** refers to the total value of additions to an economy's stock of produced assets during a given period. It includes fixed assets, change in inventories and net acquisition of valuables.

GCF can be measured using two approaches:

1. Commodity Flow Approach (Production side)
2. Fund Flow Approach (Financing side)

1. Commodity Flow Approach

- GCF is measured by adding expenditure on fixed capital formation, the change in inventories and net acquisition of valuables.

$$GCF = GFCF + CIS + NAV$$

- **(i) Gross Fixed Capital Formation (GFCF)**
  - Expenditure on the acquisition or creation of fixed assets used in production.
  - Examples: factories, buildings, machinery, equipment, roads, bridges, tools.

- **(ii) Change in Inventories or Stocks (CIS)**
  - The difference between the value of inventories at the end and at the beginning of an accounting period.
  - Inventories include raw materials, semi-finished goods and finished goods.

$$CIS = \text{Closing Stock} - \text{Opening Stock}$$

- **(iii) Net Acquisition of Valuables (NAV)**
  - The value of valuables acquired by institutional units minus the value of valuables disposed of during the period.
  - Examples: gold and precious metals (as stores of value), precious stones, paintings, works of art and antiques.

$$NAV = \text{Acquisitions} - \text{Disposals}$$

2. Fund Flow Approach

- GCF is measured from the financing side by adding gross savings generated in the domestic economy and net capital inflow from the Rest of the World (ROW).

GCF = Gross Savings + Net Capital Inflow from ROW

Common Example: GCF Using Both Approaches

A. Commodity Flow Approach

- GFCF = Rs. 500 crore
- CIS = Rs. 50 crore
- NAV = Rs. 20 crore

GCF = 500 + 50 + 20
= Rs. 570 crore

B. Fund Flow Approach

Gross Savings = Rs. 500 crore

Net Capital Inflow from ROW = Rs. 70 crore

GCF = 500 + 70
= Rs. 570 crore

Traditional (Textbook) Expenditure Method

- In the traditional approach, GDP is expressed as:

GDP = C + I + G + (X − M)

Where:

- **C** – Household Consumption Expenditure on domestically produced consumption goods and services.
- **I** – Private Investment Expenditure on domestically produced capital goods (e.g., plant, machinery, buildings).
- **G** – Government Expenditure on domestically produced consumption and capital goods and services.
- **X** – Exports of goods and services.
- **M** – Imports of goods and services.

Expanded and Alternative Forms of the Formula

- **GDP** = Private Consumption + Government Consumption + Private Investment + Government Investment + Exports − Imports
- **GDP** = Total Consumption + Total Investment + Exports − Imports

Difference between C, I and G

Component	Expenditure incurred by	Nature of Expenditure
C	Household sector	Consumption (final goods & services)
I	Private sector (firms)	Investment (capital goods)
G	Government sector	Consumption + Investment (e.g., defence, education, roads)

Why are Exports Added?

- Exports are added because they are produced within the domestic territory but purchased by the external sector (Rest of the World).

Why are Imports Subtracted?

- Imports are subtracted because GDP measures only domestic production. Imported goods and services may be included in domestic expenditure (C, I or G), but they are not produced within the domestic territory.

Key Points

- GDP under the expenditure method is measured at market prices.
- Only expenditure on final goods and services is counted.
- Expenditure on intermediate goods is excluded to avoid double counting.
- Transfer payments are not included, as they are not payments for currently produced goods or services.

Share in GDP (Approximate)

Private consumption	: around 60%
Government consumption	: around 12%
Total investment	: around 32%
Exports − Imports (Net exports)	: around 4%

Note: Net exports can be positive or negative.

SARVAN SIR ECONOMY NOTES

APPSC GROUP 1 | UPSC | TGPSC GROUP 1

COMPONENTS OF INVENTORIES

National Income Accounting | Gross Capital Formation (GCF)

1. What are inventories?

Inventories are stocks of goods held by producers, wholesalers or retailers at a particular point in time. In national income accounting, inventories generally include four components:

COMPONENT	MEANING	EXAMPLE
1. Raw materials	Materials acquired for use in the production process.	Steel, cotton, wood or other production inputs.
2. Work-in-progress	Goods that have entered production but are not yet complete.	A car that is still under assembly.
3. Finished goods	Goods completed by the manufacturer but not yet sold by that producer.	Unsold refrigerators lying in the manufacturer's warehouse.
4. Goods held for resale	Goods purchased by wholesalers or retailers for resale without further processing, but not yet sold by them.	Unsold refrigerators held in a retail shop.

2. Finished goods vs. goods held for resale

Both categories consist of goods that have completed their production process. The difference is who holds the goods and why they are holding them:

FINISHED GOODS	GOODS HELD FOR RESALE
Held by: The manufacturer. Meaning: Goods manufactured by the producer but not yet sold by that producer. Example: Refrigerators completed and waiting for sale in a factory warehouse.	Held by: A wholesaler or retailer. Meaning: Goods purchased from a manufacturer or another trader, kept for resale without further processing. Example: Refrigerators purchased by a shopkeeper but still waiting to be sold to customers.

3. Simple example: Refrigerator supply chain

Step 1: A manufacturer produces 100 refrigerators. If 20 remain unsold in the manufacturer's warehouse at the end of the accounting period, those 20 are classified as finished goods.

Step 2: A retailer purchases 80 refrigerators from the manufacturer. By the end of the period, the retailer has sold 60 and still holds 20. Those 20 are classified as goods held for resale.

Therefore, both groups are unsold finished products in an everyday sense. However, they are recorded under different inventory categories because one group is held by the producer and the other by a trader.

4. How closing inventory is related to these four components

Closing inventory means the total value of inventory remaining at the end of the accounting period. It is not another name for goods held for resale. It is the combined value of all four inventory categories:

Closing Inventory = Raw Materials + Work-in-Progress + Finished Goods + Goods Held for Resale

Example: If a business has raw materials worth ₹2 lakh, work-in-progress worth ₹1 lakh, finished goods worth ₹5 lakh and goods held for resale worth ₹3 lakh, its total closing inventory is ₹11 lakh.

5. Connection with Gross Capital Formation

The change in inventories is one of the components of Gross Capital Formation under the Commodity Flow Approach. It is measured as the difference between closing inventory and opening inventory:

Change in Inventories = Closing Inventory − Opening Inventory

6. Exam-ready summary

Remember:
• Finished goods are completed products held by the manufacturer, awaiting sale.
• Goods held for resale are products purchased by wholesalers or retailers, awaiting resale without further processing.
• Both are included in inventories, but they are separate inventory categories.
• Closing inventory is the total of all four inventory components.

SARVAN SIR ECONOMY NOTES

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GROSS CAPITAL FORMATION (GCF)

Definitions • Approaches • Examples • Classroom Clarifications

1. Definition of Gross Capital Formation

Gross Capital Formation refers to the creation or acquisition of capital assets in an economy during a given period, indicating the total investment expenditure undertaken in the country.

2. Approaches to Calculate Gross Capital Formation

GCF can be explained through two approaches:

- Commodity Flow Approach
- Fund Flow Approach

I. Commodity Flow Approach

Under the Commodity Flow Approach, Gross Capital Formation is calculated by adding Gross Fixed Capital Formation, Net Acquisition of Valuables and Change in Inventories.

COMMODITY FLOW APPROACH

$$GCF = GFCF + NAV + \Delta I$$

Where:

- GFCF – Gross Fixed Capital Formation
- NAV – Net Acquisition of Valuables
- ΔI – Change in Inventories

A. Gross Fixed Capital Formation (GFCF)

Definition: Gross Fixed Capital Formation refers to the acquisition or creation of fixed assets by an economy during a given period, which are used repeatedly or continuously in the production process.

What does it include?

- Plants and factories.
- Buildings and infrastructure.
- Machinery and equipment.
- New construction such as roads, bridges and dams.
- Intellectual property products such as software, research and development, and databases.

Example:

Particulars	Amount

Construction of factory	₹20 crore
Purchase of machinery	₹10 crore
Development of software	₹5 crore
Total GFCF	₹35 crore

GFCF = 20 + 10 + 5 = ₹35 crore

B. Net Acquisition of Valuables (NAV)

Definition: Net Acquisition of Valuables refers to the value of valuables acquired by an economy during a given period, minus the value of valuables disposed of during that period.

Valuables are assets acquired primarily as stores of value rather than for production or consumption.

Examples: gold and precious metals held as valuables; precious stones; antiques; paintings and other works of art.

INDIVIDUAL FORMULA

NAV = Acquisitions − Disposals

Example: Suppose during a financial year, acquisition of valuables is ₹100 crore and disposal of valuables is ₹40 crore.

NAV = 100 − 40 = ₹60 crore

Important Clarification: Understanding Net Acquisition of Valuables

Net acquisition is calculated at the economy level, not merely by looking at an individual buyer or seller.

1. Newly mined gold: Merely extracting gold from domestic mines does not automatically constitute Net Acquisition of Valuables. Gold can be recorded as acquisition of valuables when it is acquired and held primarily as a store of value, rather than used for production or consumption.

2. Imported gold: Merely importing gold does not automatically constitute Net Acquisition of Valuables. Its actual use and treatment must be considered. If it is acquired and held as a valuable, it can contribute to NAV. Imports must also be treated consistently in national accounting to avoid double-counting.

3. Transfer of existing gold within the domestic economy: Suppose Person A sells existing gold worth ₹40 crore to Person B. A disposes of ₹40 crore and B acquires ₹40 crore.

Net effect on the economy = ₹40 crore − ₹40 crore = Zero

A mere transfer of ownership of existing gold between domestic residents does not create an additional net acquisition of valuables for the economy.

Key concept: Extraction or production of gold, acquisition of gold, and holding gold as a valuable are distinct stages. Production or import alone does not automatically qualify as NAV.

C. Change in Inventories (ΔI)

Definition: Change in Inventories refers to the difference between the value of inventories held by producers and businesses at the end of an accounting period and their value at the beginning of that period.

Meaning of inventories: Inventories are stocks of goods held by producers and businesses for production or sale.

Components of inventories:

- Raw materials: Materials purchased for further production.
- Work-in-progress (semi-finished goods): Goods that have entered the production process but are not yet completed.
- Finished goods: Goods that are completely manufactured and ready for sale.
- Goods for resale: Goods purchased by traders for resale without further processing.
- Other inventories: Certain production supplies and materials held for use in production.

INDIVIDUAL FORMULA

ΔI = Closing Inventory − Opening Inventory

Example: A manufacturing company has opening inventory of ₹50 crore on 1 April and closing inventory of ₹80 crore on 31 March.

Particulars	Amount
Opening Inventory	₹50 crore
Closing Inventory	₹80 crore
Change in Inventories	₹30 crore

ΔI = 80 − 50 = +₹30 crore

Interpretation:

- Closing inventory greater than opening inventory → Positive change.
- Closing inventory less than opening inventory → Negative change.
- Closing inventory equal to opening inventory → Zero change.

II. Fund Flow Approach

Definition: Under the Fund Flow Approach, Gross Capital Formation is explained by identifying the sources of funds available to finance investment expenditure in an economy during a given period.

FUND FLOW APPROACH

GCF = Domestic Savings + Net Inflow

A. Domestic Savings

Domestic Savings consist of savings generated by three institutional sectors of the domestic economy:

- Household Savings
- Corporate Savings
- Government Savings

Domestic Savings = Household Savings + Corporate Savings + Government Savings

1. Household Savings

Household Savings refer to the portion of household disposable income that is not spent on consumption.

Examples: savings in bank deposits; investment in shares and securities; provident fund savings; and savings in other financial and physical assets.

2. Corporate Savings

Corporate Savings refer to the portion of corporate income retained by companies after meeting expenses and distributing dividends to shareholders.

Examples: retained earnings and undistributed corporate profits.

3. Government Savings

Government Savings refer to the surplus of government current receipts over government current expenditure.

Government Savings = Government Current Receipts − Government Current Expenditure

Examples of current receipts: tax revenue, non-tax revenue, interest receipts, and dividends and profits received by the government.

Examples of current expenditure: salaries and pensions, interest payments, subsidies, and administrative expenditure.

B. Net Inflow

Net Inflow refers, in this simplified teaching framework, to net funds received by the domestic economy from the rest of the world to help finance domestic investment. It broadly represents foreign savings or net external financing.

Examples of external financing flows include foreign direct investment (FDI), foreign portfolio investment (FPI), external commercial borrowings and other external financing flows. These are examples of financing channels; gross receipts should not automatically be added without considering repayments, outflows and national-accounting treatment.

III. Side-by-Side Comparison of GCF under Both Approaches

Illustrative values in ₹ crore. The figures are chosen to demonstrate the same total under both approaches.

Commodity Flow	Amount	Fund Flow Approach	Amount

Approach			
Gross Fixed Capital Formation (GFCF)	100	Household Savings	70
Net Acquisition of Valuables (NAV)	20	Corporate Savings	40
Change in Inventories (ΔI)	30	Government Savings	20
		Total Domestic Savings	130
		Net Inflow	20
Gross Capital Formation	150	Gross Capital Formation	150

Commodity Flow: 100 + 20 + 30 = ₹150 crore

Fund Flow: 70 + 40 + 20 + 20 = ₹150 crore

Conclusion: The total GCF is the same under both approaches in this illustrative example. The Commodity Flow Approach identifies the components of investment, while the Fund Flow Approach identifies the sources of finance in the simplified framework.

IV. Special Classroom Note: Government Capital Expenditure and Government Savings

Government contributes to capital formation by undertaking capital expenditure that creates capital assets such as dams, national highways, bridges, irrigation projects and public infrastructure.

1. Example: Government Receipts and Expenditure

Suppose the government has current receipts of ₹100 crore during a financial year and uses ₹60 crore for current expenditure such as salaries, pensions and subsidies. The remaining ₹40 crore is available as a current surplus and is used to finance capital expenditure.

Particulars	Amount
Government Current Receipts	₹100 crore
Government Current Expenditure (salaries, pensions, schemes, etc.)	₹60 crore
Surplus of current receipts over current expenditure	₹40 crore

Government Savings = 100 − 60 = ₹40 crore

Suppose the government uses ₹40 crore to construct a dam and a national highway. These are capital assets.

2. Reflection under the Commodity Flow Approach

The dam and national highway created through the ₹40 crore capital expenditure are included under Gross Fixed Capital Formation, subject to the usual national-accounting treatment.

GCF = GFCF + NAV + ΔI

Government-created fixed assets in this example: GFCF = ₹40 crore

3. Reflection under the Fund Flow Approach

Government savings do not literally mean money placed in a piggy bank or a bank account, as individuals save. In national income accounting, government savings mean the surplus of government current receipts over government current expenditure.

In this first example, current receipts are ₹100 crore and current expenditure is ₹60 crore. Therefore, government savings are ₹40 crore. The same amount is viewed as capital investment under the Commodity Flow Approach and as the government's current surplus under the Fund Flow Approach.

Commodity Flow Approach	Fund Flow Approach
₹40 crore spent on a dam and national highway; recorded under GFCF.	₹40 crore current surplus; recorded as Government Savings.

4. Important Clarification: Capital Expenditure and Government Savings Are Not Always Equal

Do not automatically treat the entire capital expenditure of the government as Government Savings. Capital expenditure may be financed partly from the government's own savings and partly through borrowings.

Consider a separate example in which the government undertakes capital expenditure of ₹40 crore, finances ₹30 crore from its own savings, and borrows the remaining ₹10 crore.

Source of Finance	Amount
Government Savings	₹30 crore
Borrowings	₹10 crore
Total financing of capital expenditure	₹40 crore

A. Commodity Flow Approach

The entire ₹40 crore expenditure creates fixed capital assets such as a dam or national highway.

GFCF = ₹40 crore

B. Fund Flow Approach

The government's own savings contribution is ₹30 crore. The remaining ₹10 crore is financed through government borrowings.

The government may borrow from banks, financial institutions or investors who purchase government securities. The funds lent to the government may ultimately be sourced from savings held by households, corporations or other financial institutions.

For this classroom illustration, the exact original saver need not be identified. The key point is that the ₹10 crore is borrowed funds made available by other economic entities; it is not Government Savings.

Commodity Flow Approach	Amount	Fund Flow / Financing Explanation	Amount
Government capital expenditure recorded under GFCF	₹40 crore	Government's own savings	₹30 crore
		Borrowed funds supplied by other sectors	₹10 crore
Total capital formation represented	₹40 crore	Total financing represented	₹40 crore

Key distinction:

- ₹40 crore = Total Government Capital Expenditure, reflected under GFCF.
- ₹30 crore = Government's own savings contribution.
- ₹10 crore = Borrowed funds made available by other economic entities; these may be sourced from household, corporate or financial-sector savings.

Remember: Borrowings are a source of finance and a liability for the government. They are not Government Savings. The underlying savings of lenders should not be counted twice as a separate addition to national savings.

V. Quick Revision: Important Formulas

COMMODITY FLOW APPROACH

$$GCF = GFCF + NAV + \Delta I$$

SIMPLIFIED FUND FLOW FRAMEWORK

$$GCF = \text{Domestic Savings} + \text{Net Inflow}$$

$$\text{Domestic Savings} = \text{Household Savings} + \text{Corporate Savings} + \text{Government Savings}$$

$$\text{Government Savings} = \text{Current Receipts} - \text{Current Expenditure}$$

$$NAV = \text{Acquisitions} - \text{Disposals}$$

$$\Delta I = \text{Closing Inventory} - \text{Opening Inventory}$$

Exam-Oriented Takeaways

- Commodity Flow Approach focuses on the components of investment expenditure.
- Fund Flow Approach focuses on the sources of finance in the simplified framework.
- GFCF includes fixed assets used repeatedly or continuously in production.
- NAV includes net acquisitions of valuables held primarily as stores of value.
- A domestic transfer of an existing valuable between residents does not create an additional net acquisition for the economy.
- Change in inventories is closing inventory minus opening inventory.
- Government savings are current receipts minus current expenditure.
- Government capital expenditure and Government Savings are different concepts.
- Borrowings finance expenditure but are not Government Savings.

SARVAN SIR ECONOMY NOTES

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GROSS CAPITAL FORMATION (GCF) - 20 ADVANCED MCQs

Medium: English Level: Moderate to Advanced Total Questions: 20

Instructions: Choose the most appropriate answer. The answer key and explanations are provided after all 20 questions.

SECTION I - COMMODITY FLOW APPROACH AND GFCF

Q1. Gross Capital Formation primarily represents:

A. Total consumption expenditure in an economy
B. Creation or acquisition of capital assets during an accounting period
C. Total government revenue receipts
D. Only expenditure incurred on infrastructure

Q2. Which of the following is the correct expression for GCF under the Commodity Flow Approach?

A. GCF = GDP - Depreciation
B. GCF = GFCF + NAV + Change in Inventories
C. GCF = Domestic Savings - Net Inflow
D. GCF = GFCF - NAV + Change in Inventories

Q3. Consider the following assets: 1. Factory building; 2. Machinery; 3. Software developed for production; 4. Gold jewellery held primarily as a store of value. Which of the above are included in GFCF?

A. 1 and 2 only
B. 1, 2 and 3 only
C. 1, 2 and 4 only
D. All four

Q4. Gross Fixed Capital Formation refers to:

A. Purchase of all financial assets
B. Creation or acquisition of fixed assets used repeatedly or continuously in production
C. Increase in the value of existing land and shares
D. Only government infrastructure expenditure

Q5. A company makes the following investments: factory construction Rs. 25 crore, machinery Rs. 15 crore and qualifying software development Rs. 10 crore. Its GFCF is:

A. Rs. 25 crore
B. Rs. 40 crore
C. Rs. 50 crore
D. Rs. 60 crore

Q6. Net Acquisition of Valuables is calculated as:

A. Disposals - Acquisitions
B. Acquisitions + Disposals
C. Acquisitions - Disposals
D. Opening stock - Closing stock

Q7. Consider the following statements about gold and Net Acquisition of Valuables: 1. Mere extraction of gold from a mine automatically constitutes NAV. 2. Gold acquired and held primarily as a store of value can qualify as a valuable. 3. Transfer of existing gold between two domestic residents automatically increases economy-wide NAV. Which of the statements are correct?

A. 1 only
B. 2 only
C. 1 and 3 only
D. 2 and 3 only

Q8. Person A sells existing gold worth Rs. 60 crore to Person B within the domestic economy. Assuming no other transactions, the net effect on economy-wide NAV is:

A. Rs. 60 crore
B. -Rs. 60 crore
C. Rs. 120 crore
D. Zero

Q9. Which of the following are generally included in inventories? 1. Raw materials; 2. Work-in-progress; 3. Finished goods; 4. Goods held for resale.

A. 1 and 2 only
B. 1, 2 and 3 only
C. 2, 3 and 4 only
D. 1, 2, 3 and 4

Q10. A firm reports opening inventory of Rs. 75 crore and closing inventory of Rs. 95 crore. The change in inventories is:

A. -Rs. 20 crore
B. Rs. 20 crore
C. Rs. 75 crore
D. Rs. 170 crore

SECTION II - FUND FLOW APPROACH

Q11. Under the Fund Flow Approach, the official financing-side expression for Gross Capital Formation in the revised national accounts is:

A. Gross Savings - Net Capital Inflow from ROW
B. Gross Savings + Net Capital Inflow from ROW
C. Gross Fixed Capital Formation + Government Revenue
D. Domestic Consumption + Net Exports

Q12. Domestic savings, under the simplified framework, consist of:

A. Household savings only
B. Government and foreign savings only
C. Household, corporate and government savings
D. Corporate savings and net exports only

Q13. Government savings are calculated as:

A. Capital receipts - Capital expenditure
B. Total receipts - Total expenditure
C. Current receipts - Current expenditure
D. Tax revenue - Borrowings

Q14. Consider the following statements: 1. Government capital expenditure can contribute to GFCF. 2. Government savings represent the surplus of current receipts over current expenditure. 3. Government capital expenditure and government savings are always equal. Which of the statements given above are correct?

A. 1 only
B. 1 and 2 only
C. 2 and 3 only
D. 1, 2 and 3

Q15. A government undertakes capital expenditure of Rs. 40 crore. It finances Rs. 30 crore from its own savings and Rs. 10 crore through borrowings. Which statement is correct?

A. GFCF is Rs. 30 crore and government savings are Rs. 40 crore.
B. GFCF is Rs. 40 crore and government savings are Rs. 30 crore.
C. GFCF is Rs. 10 crore and government savings are Rs. 30 crore.
D. Both GFCF and government savings are Rs. 40 crore.

Q16. When the government borrows Rs. 10 crore from a bank, which of the following is the most appropriate interpretation?

A. The borrowing automatically becomes government savings.
B. The borrowing is a source of finance and a liability of the government; the lender provides funds.
C. The borrowing is automatically recorded as GFCF.
D. The borrowing is government current revenue.

Q17. Consider the following statements regarding the two approaches: 1. Commodity Flow Approach identifies the components of investment. 2. Fund Flow Approach identifies the sources of financing. 3. Both approaches should provide the same GCF when measured consistently. Which of the statements given above are correct?

A. 1 only
B. 1 and 2 only
C. 2 and 3 only
D. 1, 2 and 3

SECTION III - CURRENT AFFAIRS AND NATIONAL ACCOUNTS

Q18. In February 2026, MoSPI released a new series of National Accounts Estimates with which of the following base years?

A. 2011-12
B. 2015-16
C. 2020-21
D. 2022-23

Q19. According to the official estimates for FY 2024-25 under the new National Accounts series, Gross Capital Formation at current prices was estimated at approximately:

A. Rs. 95.17 lakh crore
B. Rs. 100 lakh crore
C. Rs. 109.25 lakh crore
D. Rs. 111.13 lakh crore

Q20. According to the same official estimates for FY 2024-25, which institutional sector accounted for the largest share of Gross Savings?

A. Household sector
B. Non-financial corporations
C. General government
D. Financial corporations

ANSWER KEY AND EXPLANATIONS

Check this section after attempting all 20 questions.

1. B

Gross Capital Formation measures the creation or acquisition of capital assets during an accounting period. It represents investment in the economy, not consumption or government revenue.

2. B

Under the Commodity Flow Approach: GCF = GFCF + NAV + Change in Inventories. These are the three principal components on the expenditure or commodity side.

3. B

Factory buildings, machinery and qualifying software used in production are fixed assets and form part of GFCF. Gold held primarily as a store of value is treated under valuables, not GFCF.

4. B

GFCF covers the acquisition or creation of fixed assets intended for repeated or continuous use in production. It is not limited to government infrastructure or financial assets.

5. C

GFCF = Rs. 25 crore + Rs. 15 crore + Rs. 10 crore = Rs. 50 crore, assuming the software meets the criteria for a capitalised intellectual property product.

6. C

NAV is the value of acquisitions of valuables minus the value of disposals during the accounting period: NAV = Acquisitions - Disposals.

7. B

Only statement 2 is correct. Mining alone does not automatically create NAV; gold must be acquired and held as a store of value. A domestic transfer of existing gold creates an acquisition for one resident and a disposal for another, so it does not automatically increase economy-wide NAV.

8. D

The buyer's acquisition of Rs. 60 crore is offset by the seller's disposal of the same existing gold. With no other transaction, the net effect on economy-wide NAV is zero.

9. D

Inventories generally include raw materials, work-in-progress, finished goods and goods held for resale. They represent stocks held for use in production or sale.

10. B

Change in Inventories = Closing Inventory - Opening Inventory = Rs. 95 crore - Rs. 75 crore = Rs. 20 crore. A positive result indicates an increase in inventories.

11. B

The revised MoSPI National Accounts formulation presents the financing-side relationship as GCF = Gross Savings + Net Capital Inflow from the Rest of the World (ROW).

12. C

In the simplified domestic-savings framework used in these notes, domestic savings comprise household savings, corporate savings and government savings. The official national accounts use the defined gross-savings aggregates.

13. C

Government savings are the surplus of current receipts over current expenditure: Government Savings = Current Receipts - Current Expenditure. Borrowings are not current receipts for this purpose.

14. B

Statements 1 and 2 are correct. Government capital expenditure on qualifying fixed assets can contribute to GFCF. Government savings are measured from the current account. Capital expenditure and savings need not be equal because capital expenditure can also be financed through borrowing and other sources.

15. B

The full Rs. 40 crore spent on qualifying fixed assets contributes to GFCF. Of this, only Rs. 30 crore is financed from the government's own savings; the remaining Rs. 10 crore is financed through borrowing.

16. B

Borrowing provides funds to the government and creates a repayment liability. It is a financing source, not government savings and not itself capital formation. The use of borrowed funds may finance capital expenditure.

17. D

All three statements are correct. The commodity flow approach identifies what investment consists of; the fund flow approach identifies how it is financed. When concepts, coverage and timing are consistent, both approaches should yield the same GCF.

18. D

MoSPI's new National Accounts series released in February 2026 uses 2022-23 as the base year, replacing the earlier 2011-12 base year.

19. C

Under the new series, GCF at current prices for FY 2024-25 was estimated at approximately Rs. 109.25 lakh crore.

20. A

The household sector accounted for the largest share of gross savings in FY 2024-25, at approximately 62.1% in the cited official estimates.

Current-affairs reference: Ministry of Statistics and Programme Implementation (MoSPI), new National Accounts series with base year 2022-23, released in February 2026; official FY 2024-25 estimates. Figures are rounded as presented in the questions.

Sarvan Sir's Economy Notes

GDP Calculation – Income Method

1 Meaning / Concept

- The Income Method is a way of calculating GDP by adding all the incomes generated in the process of production within the domestic territory of a country during an accounting year.
- It measures GDP from the income side.

2 Simplified Definition

GDP under the Income Method is the sum of all factor incomes generated within the domestic territory of a country during an accounting year, plus consumption of fixed capital and net taxes.

Sarvan Sir's Economy Notes

GDP Calculation – Income Method

1 Meaning / Concept

- The Income Method is a way of calculating GDP by adding all the **incomes** generated in the process of production within the domestic territory of a country during an accounting year.

2 Simplified Definition

GDP under the Income Method is the sum of all factor incomes generated within the domestic territory of a country, plus consumption of fixed capital and net taxes (on production and products).

3 Ministry of Statistics and Programme Implementation (MoSPI) Formula

$$GDP_{MP} = CE + OS/MI + CFC + NTP + NTPR$$

Where:

- **CE** = Compensation of Employees
- **OS/MI** = Operating Surplus / Mixed Income
- **CFC** = Consumption of Fixed Capital (Depreciation)
- **NTP** = Net Taxes on Production (Taxes on production – Subsidies on production)
- **NTPR** = Net Taxes on Products (Taxes on products – Subsidies on products)

4 Illustration (₹ crore)

Particulars	Amount (₹ crore)
Compensation of Employees (CE)	500
Operating Surplus / Mixed Income (OS/MI)	250
Consumption of Fixed Capital (CFC)	100
Net Taxes on Production (NTP)	50
GVA at Basic Prices (GVA_{BP})	**900**
Net Taxes on Products (NTPR)	100
GDP at Market Prices (GDP_{MP})	**1,000**

Calculation:

- $GVA_{BP} = CE + OS/MI + CFC + NTP$
  $= 500 + 250 + 100 + 50$
  $=$ ₹ 900 crore
- $GDP_{MP} = GVA_{BP} + NTPR$
  $= 900 + 100$
  $=$ ₹ 1,000 crore

GDP = ₹ 1,000 crore

Sarvan Sir's Economy Notes

GDP Calculation – Income Method : Key Points & Additions

1 Factor Income vs GDP at Market Prices

- The Income Method begins with factor incomes (CE, OS, MI).
- To get GDP at Market Prices, we add **CFC** and **net taxes** (on production and products).

Formula (MoSPI)

$$GVA_{BP} = CE + OS/MI + CFC + NTP$$

$$\Downarrow$$

$$GDP_{MP} = GVA_{BP} + NTPR$$

2 Gross vs Net Operating Surplus

- **Gross Operating Surplus (GOS)** : Includes consumption of fixed capital (CFC).
- **Net Operating Surplus (NOS)** : Excludes CFC.

3 Mixed Income (MI)

- Income earned by self-employed persons and unincorporated enterprises (where it is difficult to separate wages, rent, interest and profit).
- **Example :** A small shopkeeper, a self-employed farmer, a local artisan, etc.

4 Limitations of Income Method

- Difficulty in estimating income from the **informal sector** (unorganised sector).
- Possibility of **under-reporting** of income (e.g., in self-employment, small businesses).
- Difficulty in separating **mixed income** into wages, rent, interest and profit.

Key Takeaway

- Factor incomes = CE + OS + MI.
- CFC and taxes are not factor incomes.
- They are **valuation adjustments** needed to arrive at GDP at Market Prices.

Methods of Calculating GDP – At a Glance

Method	What does it measure?	Key Expression (Idea)
Value Added Method	Value created in production	$\sum GVA_{BP} + NTPR$
Income Method	Income generated in production	$CE + OS/MI + CFC + NTP + NTPR$
Expenditure Method	Expenditure incurred on final goods and services	$PFCE + GFCE + GCF + (X - M)$

SARVAN SIR

ECONOMY NOTES

NATIONAL INCOME ACCOUNTING

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TOPICS TO BE COVERED – INDEX

No.	Topic
01	GVA at Factor Cost
02	GNP – Gross National Product
03	GDI – Gross Domestic Income
04	GNI – Gross National Income
05	NDP – Net Domestic Product
06	NNI – Net National Income
07	Traditional National Income vs Current MoSPI Presentation
08	Factor Income
09	Primary Income and Secondary Income
10	Personal Income
11	Personal Disposable Income
12	GNDI – Gross National Disposable Income
13	NNDI – Net National Disposable Income
14	GNDI and Final Consumption / Gross Saving
15	Complete Formula Revision Sheet

Learning sequence: First understand production and national income aggregates; then distinguish primary income from current transfers; finally connect these ideas to disposable income and GNDI.

PART I – IMPORTANT NATIONAL INCOME AGGREGATES

1. GVA at Factor Cost

Definition: Gross Value Added at Factor Cost refers to value added through production after deducting intermediate consumption and net other taxes on production from GVA at Basic Prices.

GVA at Factor Cost = GVA at Basic Prices − Net Other Taxes on Production

Shortcut: $GVA_{FC} = GVA_{BP} - NITP$, where NITP means Net Other Taxes on Production.

Components	Amount (₹ crore)
Compensation of Employees	40
Operating Surplus	20
Mixed Income	25
Consumption of Fixed Capital (Depreciation)	15
Net Other Taxes on Production	20
GVA at Basic Prices	120

GVA at Factor Cost = 120 − 20 = ₹100 crore

Why is depreciation included? GVA at Factor Cost is a gross measure. It includes Consumption of Fixed Capital (CFC), which represents the loss in value of fixed assets used during production.

NVA at Factor Cost = GVA at Factor Cost − CFC

2. GNP – Gross National Product

Definition: GNP is the gross domestic product adjusted for net primary income from the Rest of the World (ROW).

GNP = GDP + NFIA

Expanded formula: GNP = GDP + Primary Income Received from ROW − Primary Income Paid to ROW. NFIA is the traditional term Net Factor Income from Abroad.

Factor of Production	Traditional factor income
Land	Rent
Labour	Wages
Capital	Interest
Entrepreneurship	Profit

Primary income received from ROW	Primary income paid to ROW
Compensation received by Indian resident units from abroad	Compensation paid to non-resident employees in India
Interest received from foreign investments	Interest paid to foreign lenders

Primary income received from ROW	Primary income paid to ROW
Dividends received from abroad	Dividends paid to foreign investors
Investment income received by Indian residents	Investment income accruing to foreign investors in India

Numerical example: GDP = ₹1,000 crore; primary income received from ROW = ₹150 crore; primary income paid to ROW = ₹100 crore.

GNP = 1,000 + (150 − 100) = ₹1,050 crore

- Economic residence, not citizenship, determines whether a unit is resident or non-resident.
- Personal remittances are current transfers, not primary income; they are not included in NFIA.

3. GDI – Gross Domestic Income

Definition: GDI refers to the income generated through the production of goods and services within the domestic territory during a given period.

- GDP measures production; GDI measures the income generated through that production.

GDP = GDI (conceptually)

Illustrative income component	Amount (₹ crore)
Compensation of Employees	40
Gross Operating Surplus	25
Gross Mixed Income	15
Consumption of Fixed Capital	10
Net taxes on production and products	10
GDI	100

If GDP is ₹100 crore, GDI is conceptually ₹100 crore. Actual statistical estimates may differ because of statistical discrepancies arising from different data sources and estimation methods.

Terminology caution: GDI may also mean Gross Disposable Income in some national-accounts contexts. Interpret it according to the context.

4. GNI – Gross National Income

Definition: GNI refers to the total gross primary income earned by resident institutional units during a given period, including net primary income from the ROW.

GNI = GDP + Net Primary Income from ROW

GNI = GDP + Primary Income Received from ROW − Primary Income Paid to ROW

Relationship: GNP and GNI are corresponding national aggregates; GNI is the modern income-oriented terminology.

5. NDP – Net Domestic Product

Definition: NDP is the value of final goods and services produced within the domestic territory during a given period, after deducting Consumption of Fixed Capital (depreciation).

NDP = GDP − Depreciation = GDP − CFC

Component	Amount (₹ crore)
GDP	1,000
Depreciation / CFC	100
NDP	900

NDP = 1,000 − 100 = ₹900 crore

- GDP is a gross domestic measure.
- NDP is a net domestic measure.
- Domestic refers to production within the domestic territory.

6. NNI – Net National Income

Definition: NNI refers to the total net primary income earned by resident institutional units during a given period. In simple terms, NNI is GNI after deducting CFC.

NNI = GNI − Depreciation = GNI − CFC

Component	Amount (₹ crore)
GNI at Market Prices	1,050
Depreciation / CFC	100
NNI at Market Prices	950

NNI = 1,050 − 100 = ₹950 crore

NNI = NDP + Net Primary Income from ROW

7. Traditional National Income vs Current MoSPI Presentation

Traditional textbook convention: National Income is conventionally defined as NNP at Factor Cost.

Traditional National Income = NNP at Factor Cost

NNP at Factor Cost = NNP at Market Prices − Net Indirect Taxes

Current national-accounts presentation: MoSPI reports Net National Income (NNI) at Market Prices as a national-accounts aggregate. NNI at Market Prices is equivalent to NNP at Market Prices.

NNI at Market Prices = GNI at Market Prices − CFC

Component	Amount (₹ crore)
GNI at Market Prices	1,050
Depreciation / CFC	100
NNI at Market Prices	950
Net Indirect Taxes (illustrative)	80
NNP at Factor Cost (traditional measure)	870

EXAM DISTINCTION: Traditional textbook definition – NNP at Factor Cost. Current MoSPI national-accounts aggregate – NNI at Market Prices (equivalent to NNP at Market Prices). Do not treat these as the same valuation.

PART II – UNDERSTANDING DISPOSABLE INCOME

8. Factor Income

Definition: Factor Income refers to income received by the owners of factors of production for providing their factors in the production process.

Factor of Production	Factor Income	Example
Land	Rent	Rent received by a landlord
Labour	Wages	Salary received by an employee
Capital	Interest	Interest received by a lender
Entrepreneurship	Profit	Profit earned by an entrepreneur

9. Primary Income and Secondary Income

A. Primary Income

Primary income refers to income accruing to institutional units as a result of their participation in production or ownership of assets used in production.

- Compensation of employees (wages and salaries).
- Interest received.
- Dividends received.
- Rent from natural resources.
- Other investment income.

B. Secondary Income – Current Transfers

Secondary income refers to transfers received without providing goods, services or factors of production in return. These are commonly described as current transfers.

Category	Examples
Government current transfers	Social security pensions, eligible welfare benefits, scholarships and cash assistance schemes
Personal transfers	Remittances received from relatives living abroad
Gifts and donations	Money received as gifts or donations
Current grants	Current grants received from foreign governments or international organisations
Other current transfers	Certain insurance-related current transfers and other eligible transfers

Primary Income	Secondary Income
Arises from production or ownership of assets	Arises from transfers
Wages, interest, rent, dividends	Remittances, gifts, eligible grants and benefits
A primary-income earning relationship exists	No corresponding productive contribution is required

Important: Primary income and secondary income are separate categories. Current transfers are not part of primary income.

10. Personal Income (PI)

Definition: Personal Income refers to income received by individuals or households from primary income and current transfers, subject to relevant national-accounts adjustments.

Simplified teaching relationship: PI = Primary Income Received + Current Transfers Received

Particulars	Amount
Primary / factor income received	₹1,000
Current transfers received	₹500
Personal Income	₹1,500

11. Personal Disposable Income (PDI)

Definition: Personal Disposable Income refers to the income available to individuals or households for consumption and saving after deducting relevant current taxes and social contributions.

PDI = Personal Income − Personal Current Taxes − Relevant Social Contributions

Social contributions are payments made towards social insurance and social security schemes. Examples include employee contributions to EPF, ESI and qualifying pension or social insurance schemes.

- An employee's EPF contribution is a social contribution.
- An employee's ESI contribution, where applicable, is a social contribution.
- Contributions to qualifying social insurance or pension schemes may be social contributions.
- An ordinary private life or health insurance premium is not automatically a social contribution.

Particulars	Amount
Primary / factor income received	₹1,000
Current transfers received	₹500
Personal Income	₹1,500
Personal income tax	₹200
EPF contribution (illustrative)	₹100
Qualifying social insurance contribution (illustrative)	₹100
Qualifying pension contribution (illustrative)	₹100
Total deductions	₹500
Personal Disposable Income	₹1,000

PDI = 1,500 − 200 − 300 = ₹1,000

Utilisation of PDI: Personal Disposable Income is used for consumption and saving.

PDI = Consumption + Saving

Particulars	Amount
Personal Disposable Income	₹1,000
Consumption expenditure	₹700
Personal saving	₹300

The illustrative deductions assume that the listed ₹300 are qualifying social contributions. Not every insurance premium qualifies. Formal national-accounts treatment is more detailed than a simple payslip calculation.

PART III – GROSS NATIONAL DISPOSABLE INCOME

12. GNDI – Gross National Disposable Income

Definition: Gross National Disposable Income refers to the total income available to resident institutional units of a country for consumption and saving during a given period, after accounting for net current transfers from the Rest of the World.

Why do we need GNDI after GNI?

GNI	GNDI
Measures gross primary income of resident institutional units	Measures gross income available after net current transfers
Includes net primary income from ROW	Includes net primary income and net current transfers from ROW
Excludes other current transfers	Includes other current transfers

GNDI = GNI + Net Current Transfers from ROW

GNDI = GNI + Current Transfers Received from ROW − Current Transfers Paid to ROW

Numerical example:

Component	Amount (₹ crore)
GNI	1,050
Current transfers received from ROW	80
Current transfers paid to ROW	30
Net current transfers	50
GNDI	1,100

GNDI = 1,050 + (80 − 30) = ₹1,100 crore

13. NNDI – Net National Disposable Income

Definition: NNDI refers to Gross National Disposable Income after deducting Consumption of Fixed Capital.

NNDI = NNI + Net Current Transfers from ROW

NNDI = GNDI − CFC

GNDI = NNDI + CFC

14. GNDI and Final Consumption / Gross Saving

At the level of the entire national economy, gross national disposable income is used for final consumption or retained as gross saving.

GNDI = Final Consumption Expenditure + Gross Saving

Domestic transfers between Indian resident units do not, by themselves, increase GNDI. At the national level, domestic transfers cancel out when the economy is consolidated. Net current transfers from the ROW affect GNDI.

PART IV – QUICK REVISION

15. Complete Formula Revision Sheet

Aggregate	Formula / Relationship
GVA at Factor Cost	GVA at Basic Prices − Net Other Taxes on Production
GNP	GDP + NFIA
GDI	Conceptually equal to GDP
GNI	GDP + Net Primary Income from ROW
NDP	GDP − CFC
NNI	GNI − CFC
NNP at Market Prices	GNP at Market Prices − CFC
NNP at Factor Cost	NNP at Market Prices − Net Indirect Taxes
Traditional National Income	NNP at Factor Cost
Current MoSPI NNI aggregate	NNI at Market Prices
Personal Income (simplified)	Primary Income Received + Current Transfers Received
Personal Disposable Income	PI − Relevant Current Taxes − Relevant Social Contributions
GNDI	GNI + Net Current Transfers from ROW
NNDI	NNI + Net Current Transfers from ROW
GNDI	NNDI + CFC
Use of GNDI	Final Consumption Expenditure + Gross Saving

Key Examination Takeaways

- Factor income is earned by providing factors of production: rent, wages, interest and profit.
- Primary income arises from production or ownership of assets.
- Secondary income consists of current transfers; remittances are current transfers, not primary income.
- Personal Income is a household-level concept; GNDI is a national-economy-level concept.
- Personal Disposable Income is available for consumption and saving after relevant deductions.
- GNI includes net primary income from ROW; GNDI additionally includes net current transfers from ROW.
- NDP = GDP − CFC; NNI = GNI − CFC.
- Traditional textbook National Income is NNP at Factor Cost; MoSPI's current national-accounts presentation reports NNI at Market Prices.

END OF CURRENT NOTES – National Income Accounting: From GVA and GDP to GNDI

SARVAN SIR ECONOMY NOTES

NATIONAL INCOME ACCOUNTING

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Instructions: Choose the most appropriate answer. In statement-based questions, select the option that correctly identifies the valid statements.

QUESTIONS

Each question is shown separately, followed by its statements (where applicable) and then the four answer choices.

1. Consider the following statements about Gross Value Added at Factor Cost (GVA-FC):

1. GVA-FC is obtained by deducting net other taxes on production from GVA at Basic Prices.
2. GVA-FC includes consumption of fixed capital because it is a gross measure.
3. GVA-FC is the principal valuation measure used in India's current national accounts.

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

2. If GVA at Basic Prices is Rs. 800 crore and net other taxes on production are Rs. 60 crore, the GVA at Factor Cost is:

A. Rs. 740 crore

B. Rs. 860 crore

C. Rs. 800 crore

D. Rs. 680 crore

3. Which of the following are classified as primary income received from the Rest of the World by residents of India?

1. Interest received by an Indian resident on an investment in a foreign company.
2. Personal remittances received by an Indian household from a relative working abroad.
3. Dividends received by an Indian resident from foreign investments.

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

4. If India's GDP is Rs. 2,000 crore, primary income received from the Rest of the World is Rs. 250 crore and primary income paid to the Rest of the World is Rs. 300 crore, India's GNP is:

A. Rs. 2,250 crore

B. Rs. 1,950 crore

C. Rs. 2,050 crore

D. Rs. 1,700 crore

5. Consider the following statements regarding GDP and Gross Domestic Income (GDI):

1. GDP and GDI are conceptually equal because production generates an equivalent amount of income.
2. In actual national accounts estimates, GDP and GDI may differ due to statistical discrepancies.

A. 1 only

B. Both 1 and 2

C. 2 only

D. Neither 1 nor 2

6. Consider the following statements about Gross National Income (GNI):

1. GNI is obtained by adding net primary income from the Rest of the World to GDP.
2. Personal remittances received from relatives abroad are included in net primary income from the Rest of the World.
3. GNI is an income-based measure of the income earned by resident institutional units.

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

7. If GDP is Rs. 5,000 crore and consumption of fixed capital is Rs. 400 crore, NDP is:

A. Rs. 5,400 crore

B. Rs. 4,600 crore

C. Rs. 5,000 crore

D. Rs. 4,400 crore

8. GNI at Market Prices is Rs. 4,500 crore. Net primary income from the Rest of the World is Rs. 200 crore and consumption of fixed capital is Rs. 500 crore. What is NDP at Market Prices?

A. Rs. 4,500 crore

B. Rs. 4,000 crore

C. Rs. 3,800 crore

D. Rs. 3,500 crore

9. According to the traditional Indian textbook convention, National Income is defined as:

A. GDP at Market Prices

B. GNP at Market Prices

C. NNP at Factor Cost

D. NDP at Factor Cost

10. Which of the following correctly describes the current MoSPI presentation of the net national income aggregate?

A. NNI at Factor Cost is the principal aggregate and is identical to GDP at Market Prices.

B. NNI at Market Prices is reported and is equivalent to NNP at Market Prices.

C. NNI excludes consumption of fixed capital.

D. NNI at Market Prices is equivalent to GVA at Basic Prices.

11. Consider the following traditional factor-income relationships:

1. Rent is the return to land.
2. Wages are the return to labour.
3. Interest is the return to capital.
4. Profit is the return to entrepreneurship.

A. 1 and 2 only

B. 2 and 3 only

C. 1, 2 and 3 only

D. 1, 2, 3 and 4

12. Which of the following is correctly classified as secondary income (a current transfer)?

A. Interest received on a foreign investment

B. Wages received by an Indian resident working abroad

C. Personal remittances received from a relative abroad

D. Dividends received from shares held in a foreign company

13. Consider the following statements about current transfers:

1. Current transfers may be received from governments.
2. Personal remittances may be classified as current transfers.
3. All current transfers received by Indian residents necessarily increase India's Gross National Disposable Income.

A. 1 and 2 only

B. 2 and 3 only

C. 1 and 3 only

D. 1, 2 and 3

14. An individual receives primary income of Rs. 80,000 and current transfers of Rs. 20,000. The individual pays personal current taxes of Rs. 10,000 and qualifying social contributions of Rs. 5,000. What is the individual's Personal Disposable Income (PDI)?

A. Rs. 1,00,000

B. Rs. 90,000

C. Rs. 85,000

D. Rs. 75,000

15. Which of the following are generally treated as social contributions in national accounting?

1. An employee's contribution to the Employees' Provident Fund (EPF).
2. An employer's contribution to the Employees' Provident Fund (EPF).
3. A premium paid by an individual for an ordinary private life-insurance policy.

A. 1 only

B. 1 and 2 only

C. 2 and 3 only

D. 1, 2 and 3

16. Consider the following statements about Personal Disposable Income (PDI):

1. PDI is available to individuals or households for consumption and saving.

2. PDI is derived, in a simplified presentation, after deducting relevant personal current taxes and social contributions from Personal Income.

3. PDI and Gross National Disposable Income (GNDI) measure disposable income at the same institutional-sector level.

A. 1 only

B. 1 and 2 only

C. 2 and 3 only

D. 1, 2 and 3

17. India's GNI is Rs. 3,000 crore. Current transfers received from the Rest of the World are Rs. 180 crore and current transfers paid to the Rest of the World are Rs. 80 crore. India's GNDI is:

A. Rs. 3,100 crore

B. Rs. 3,180 crore

C. Rs. 2,900 crore

D. Rs. 3,260 crore

18. Consider the following statements regarding Gross National Disposable Income (GNDI):

1. GNDI includes net primary income from the Rest of the World through GNI.

2. GNDI includes net current transfers from the Rest of the World.

3. At the national-economy level, GNDI is equal to final consumption expenditure plus gross saving.

A. 1 only

B. 1 and 2 only

C. 2 and 3 only

D. 1, 2 and 3

19. If GNDI is Rs. 2,500 crore and consumption of fixed capital is Rs. 200 crore, Net National Disposable Income (NNDI) is:

A. Rs. 2,700 crore

B. Rs. 2,300 crore

C. Rs. 2,500 crore

D. Rs. 2,100 crore

20. Consider the following relationships in national income accounting:

1. NDP = GDP minus consumption of fixed capital.

2. NNI = GNI minus consumption of fixed capital.

3. NNDI = NNI plus net current transfers from the Rest of the World.

4. GNDI = NNDI plus consumption of fixed capital.

A. 1 and 2 only

B. 2 and 3 only

C. 1, 2 and 4 only

D. 1, 2, 3 and 4

ANSWER KEY AND EXPLANATIONS

Attempt all questions before referring to this section.

Question 1: A — 1 and 2 only

GVA-FC = GVA at Basic Prices minus net other taxes on production. It includes CFC because it is a gross measure. India's current national accounts primarily present GVA at Basic Prices, not GVA-FC.

Question 2: A — Rs. 740 crore

GVA-FC = Rs. 800 crore minus Rs. 60 crore = Rs. 740 crore.

Question 3: C — 1 and 3 only

Interest and dividends received from foreign investments are primary income. Personal remittances are current transfers (secondary income), not primary income.

Question 4: B — Rs. 1,950 crore

GNP = GDP + net primary income from the Rest of the World = 2,000 + (250 minus 300) = Rs. 1,950 crore.

Question 5: B — Both 1 and 2

GDP measures domestic production while GDI measures the income generated by that production. They are conceptually equal, although actual estimates may differ because of statistical discrepancies.

Question 6: C — 1 and 3 only

GNI = GDP + net primary income from the Rest of the World. Remittances are current transfers, not primary income. GNI captures primary income earned by resident institutional units.

Question 7: B — Rs. 4,600 crore

NDP = GDP minus CFC = 5,000 minus 400 = Rs. 4,600 crore.

Question 8: C — Rs. 3,800 crore

GDP = GNI minus net primary income from ROW = 4,500 minus 200 = Rs. 4,300 crore. NDP = GDP minus CFC = 4,300 minus 500 = Rs. 3,800 crore.

Question 9: C — NNP at Factor Cost

Under the traditional Indian textbook convention, National Income is NNP at Factor Cost. Distinguish this from the current MoSPI presentation of NNI at Market Prices.

Question 10: B — NNI at Market Prices is equivalent to NNP at Market Prices

NNI at Market Prices = GNI at Market Prices minus CFC. It is equivalent to NNP at Market Prices, not GVA at Basic Prices.

Question 11: D — 1, 2, 3 and 4

Traditional factor-income classification: land—rent; labour—wages; capital—interest; entrepreneurship—profit.

Question 12: C — Personal remittances received from a relative abroad

Remittances are transfers without a corresponding factor service or production contribution. Interest, wages and dividends are primary income.

Question 13: A — 1 and 2 only

Current transfers can be received from governments, and personal remittances can be current transfers. GNDI uses net current transfers from ROW. Transfers paid abroad reduce it; domestic transfers between residents cancel at the national level.

Question 14: C — Rs. 85,000

Personal Income = Rs. 80,000 + Rs. 20,000 = Rs. 1,00,000. PDI = Rs. 1,00,000 minus Rs. 10,000 minus Rs. 5,000 = Rs. 85,000. This is a simplified classroom relationship.

Question 15: B — 1 and 2 only

Employee and employer contributions to recognised social insurance/provident fund arrangements are social contributions. An ordinary private life-insurance premium is not automatically a social contribution.

Question 16: B — 1 and 2 only

PDI is a personal/household-sector concept; GNDI is a whole-economy concept. They concern disposable income but do not measure it at the same institutional-sector level and are not identical aggregates.

Question 17: A — Rs. 3,100 crore

GNDI = GNI + net current transfers from ROW = 3,000 + (180 minus 80) = Rs. 3,100 crore.

Question 18: D — 1, 2 and 3

GNDI incorporates net primary income through GNI and net current transfers from ROW. At the whole-economy level, disposable income is allocated between final consumption expenditure and gross saving.

Question 19: B — Rs. 2,300 crore

NNDI = GNDI minus CFC = 2,500 minus 200 = Rs. 2,300 crore.

Question 20: D — 1, 2, 3 and 4

NDP and NNI are net aggregates obtained by deducting CFC from GDP and GNI respectively. NNDI = NNI + net current transfers from ROW, and GNDI = NNDI + CFC.

QUICK REVISION

GVA-FC = GVA at Basic Prices minus Net Other Taxes on Production.

GNI = GDP + Net Primary Income from the Rest of the World.

GNDI = GNI + Net Current Transfers from the Rest of the World.

NNDI = GNDI minus Consumption of Fixed Capital.

PDI is a personal/household-sector aggregate; GNDI is a whole-economy aggregate.

Personal remittances are current transfers, not primary income.

SARVAN SIR

ECONOMY NOTES

NATIONAL INCOME ACCOUNTING

PART II | STUDENT EDITION

ABOUT THESE NOTES A simple, concept-wise guide to selected measures of disposable income, with clear formulas and illustrative examples.

Net National Disposable Income (NNDI)

Definition

Net National Disposable Income is the income available to the residents of a country for consumption and saving after deducting depreciation from Gross National Disposable Income.

Formula

$$NNDI = GNDI - CFC$$

GNDI – Gross National Disposable Income | CFC – Consumption of Fixed Capital (Depreciation)

Illustrative Example

Particulars	Amount
Gross National Disposable Income (GNDI)	₹5,000 crore
Consumption of Fixed Capital (CFC)	₹400 crore
Net National Disposable Income (NNDI)	₹4,600 crore

Calculation: ₹5,000 crore − ₹400 crore

$$NNDI = ₹4,600 \text{ crore}$$

Remember

$$GNDI = NNDI + CFC$$

Gross Disposable Income of Government (GDIG)

Formula

GDIG = GFCE + Gross Savings of Government

GFCE – Government Final Consumption Expenditure

What is included in Government Final Consumption Expenditure (GFCE)?

Government spends money and uses resources to provide public services. The main components considered in this classroom breakdown are:

Component	Examples	Amount
1. Compensation of government employees	Salaries, wages and related employee benefits.	₹500 crore
2. Intermediate consumption	Electricity, stationery, fuel, office supplies and maintenance.	₹200 crore
3. Consumption of Fixed Capital (CFC)	Depreciation of government buildings, vehicles, machinery and equipment.	₹100 crore
4. Goods and services provided to households	Government-purchased medical, educational and similar services supplied to people.	₹200 crore
Total GFCE		₹1,000 crore

Why is CFC included in GFCE?

Government uses buildings, vehicles, machinery and equipment to provide public services. These assets gradually wear out. CFC represents the value of the fixed assets used up while providing those services. Therefore, it is included in the cost of government services.

REMEMBER CFC is a cost recognised in national accounting. It does not necessarily mean that the government made a cash payment for depreciation.

What is Gross Savings of Government?

Gross savings are the part of the government's disposable income that remains after meeting its final consumption expenditure.

Gross Savings = GDIG − GFCE

Illustrative Example

Particulars	Amount
Government Final Consumption Expenditure (GFCE)	₹1,000 crore
Gross Savings of Government	₹300 crore
Gross Disposable Income of Government (GDIG)	₹1,300 crore

Calculation: GDIG = ₹1,000 crore + ₹300 crore

GDIG = ₹1,300 crore

Gross Disposable Income of Households (GDIH)

Formula

GDIH = HFCE + Gross Savings of Households

HFCE – Household Final Consumption Expenditure

What is included in Household Final Consumption Expenditure (HFCE)?

It includes household expenditure on goods and services for personal consumption.

Component	Illustrative amount
1. Food and beverages	₹300 crore
2. Clothing and footwear	₹100 crore
3. Housing, electricity and water	₹150 crore
4. Transport	₹100 crore
5. Healthcare	₹80 crore
6. Education	₹70 crore
7. Communication, recreation and other services	₹100 crore
Total HFCE	₹900 crore

What is Gross Savings of Households?

Gross savings are the part of household disposable income that remains after meeting household final consumption expenditure.

- Household savings may be held as bank deposits.
- They may be invested in financial assets or held in other forms.

Illustrative Example

Particulars	Amount
Household Final Consumption Expenditure (HFCE)	₹900 crore
Gross Savings of Households	₹300 crore
Gross Disposable Income of Households (GDIH)	₹1,200 crore

Calculation: GDIH = ₹900 crore + ₹300 crore

GDIH = ₹1,200 crore

Gross Disposable Income of Corporate Sector

Formula / Relationship

Gross Disposable Income of Corporations = Gross Savings of Corporations

Corporations do not have final consumption expenditure in the same way as households and government. Therefore, their gross disposable income is equal to their gross savings.

How is corporate expenditure classified?

Corporate expenditure	How it is classified
Raw materials, electricity, fuel and other inputs used in production	Intermediate Consumption
Machinery, equipment, buildings and other fixed assets	Investment / Gross Fixed Capital Formation (GFCF)

IMPORTANT Expenditure on raw materials and other production inputs is not final consumption expenditure. Purchases of machinery and buildings are investment expenditure, not savings.

What is Gross Savings of Corporations?

Gross savings represent the income retained by corporations. They include retained earnings and the amount set aside to account for the use of fixed assets (CFC).

Illustrative Example

Particulars	Amount
Retained earnings / net savings	₹350 crore
Consumption of Fixed Capital (CFC)	₹150 crore
Gross Savings of Corporations	₹500 crore
Gross Disposable Income of Corporations	₹500 crore

Calculation: Gross Savings = ₹350 crore + ₹150 crore

Gross Disposable Income = ₹500 crore

REMEMBER Corporate investment in machinery, buildings and equipment is recorded separately as capital formation. It is not itself corporate savings.

QUICK REVISION

Concept	Formula
1. Net National Disposable Income	NNDI = GNDI − CFC
2. Government Disposable Income	GDIG = GFCE + Government Gross Savings
3. Household Disposable Income	GDIH = HFCE + Household Gross Savings
4. Corporate Disposable Income	Gross Disposable Income of Corporations = Corporate Gross Savings

NOTE TO STUDENTS All amounts in the examples are illustrative and are used only to explain the concepts.

NATIONAL INCOME ACCOUNTING

Comprehensive Class Notes

GDP • Real and Nominal Growth • COR • ICOR

Sarvan Sir Economy Notes

Scope: Consolidated notes covering the concepts discussed in class, with worked examples and exam-oriented interpretations.

1. Nominal GDP (GDP at Current Prices)

Definition: Nominal GDP is the total monetary value of all final goods and services produced within the domestic territory of a country during a given period, measured at the prices prevailing in that same period.

Nominal GDP = Current Year Prices × Current Year Quantities of Goods and Services Produced

Example – rice production: Assume 100 units are produced in both years.

Year	Quantity produced	Price per unit	Nominal GDP
2025	100 units	₹10	₹1,000
2026	100 units	₹12	₹1,200

Explanation: Production remains unchanged at 100 units. Nominal GDP rises from ₹1,000 to ₹1,200 because the price rises from ₹10 to ₹12.

Key point: Nominal GDP reflects the combined effect of changes in prices and quantities. An increase in nominal GDP does not necessarily mean real production has increased.

2. Real GDP (GDP at Constant Prices)

Definition: Real GDP is the total monetary value of final goods and services produced within the domestic territory during a given period, measured using the prices of a selected base year.

Real GDP = Base Year Prices × Current Year Quantities of Goods and Services Produced

Example – using 2025 as the base year:

Year	Quantity	Actual price	Base-year price	Real GDP
2025	100 units	₹10	₹10	₹1,000
2026	100 units	₹12	₹10	₹1,000

Explanation: Although the actual price rises to ₹12 in 2026, the calculation uses the base-year price of ₹10. Since quantity remains 100 units, real GDP remains ₹1,000.

Key point: Real GDP removes the effect of price changes and is used to assess changes in the volume of production.

3. Base Year

Definition: A base year is the reference year whose prices are used to calculate GDP at constant prices.

Characteristics of an ideal base year:

1. It should be a normal year without abnormal economic fluctuations.
2. It should not be marked by extraordinary price fluctuations in goods and services.
3. It should represent relatively stable and normal economic conditions.
4. It should have reliable, comprehensive and reasonably up-to-date statistical data.

Clarification: An ideal base year does not mean prices must remain unchanged throughout that year. Rather, it should provide a representative and relatively undistorted reference for comparison.

India's National Accounts base year: MoSPI (Ministry of Statistics and Programme Implementation) compiles India's National Accounts Statistics. The earlier base year was 2011–12. The revised National Accounts series released on 27 February 2026 adopted 2022–23 as the new base year.

Why revise the base year? Revision helps incorporate more recent economic structures, updated data sources and improved statistical methods, so national accounts better reflect the economy as it exists today.

4. National Income Growth at Current Prices

Definition: Percentage increase in national income measured at prices prevailing in the respective years. It is also called nominal national income growth.

Example:

Year	National income at current prices
2025	₹1,000 crore
2026	₹1,200 crore

$$\text{Growth Rate} = \frac{\text{Current Year NI} - \text{Previous Year NI}}{\text{Previous Year NI}} \times 100$$

Calculation: [(1,200 − 1,000) ÷ 1,000] × 100 = 20%.

Interpretation: National income increased by 20% at current prices. This reflects the combined effect of output and price changes; it does not remove inflation.

5. National Income Growth at Constant Prices

Definition: Percentage increase in national income measured at the prices of a selected base year. It is also called real national income growth.

Example:

Year	National income at constant prices
2025	₹1,000 crore
2026	₹1,050 crore

$$\text{Growth Rate} = \frac{\text{Current Year NI} - \text{Previous Year NI}}{\text{Previous Year NI}} \times 100$$

Calculation: [(1,050 − 1,000) ÷ 1,000] × 100 = 5%.

Interpretation: National income increased by 5% in real terms because constant prices remove the price-change effect.

Important distinction: In the traditional Indian national income concept, national income is measured as Net National Product at Factor Cost (NNP at FC). GDP and national income are not identical. GDP growth and national income growth should not be used interchangeably without specifying the measure.

6. Capital-Output Ratio (COR)

Definition: COR indicates the relationship between total capital employed and the total value of output generated.

$$\text{COR} = \frac{\text{Total Capital Employed}}{\text{Total Value of Output}}$$

Example: A factory is established with capital of ₹100. It produces 5 goods, each worth ₹1. Total value of output is ₹5.

Particular	Value
Total capital employed	₹100
Quantity produced	5 units
Value per unit	₹1
Total value of output	₹5

Calculation: COR = 100 ÷ 5 = 20.

Interpretation: ₹20 of capital is employed to generate ₹1 worth of output. COR can be expressed as 20:1.

Note: Monetary value of output is used rather than physical quantity alone, so capital and output are compared in consistent value terms.

7. Incremental Capital-Output Ratio (ICOR)

Definition: ICOR indicates the amount of additional capital investment required to generate one additional unit of output.

$$\text{ICOR} = \frac{\text{Additional Capital Investment}}{\text{Additional Output}}$$

Worked example – factory expansion: Initially, a factory has capital of ₹100 and produces 5 goods worth ₹1 each. It invests an additional ₹50 in machinery, enabling it to produce 5 additional goods, each worth ₹1.

Particular	Before expansion	After expansion
Total capital	₹100	₹150
Total output quantity	5 units	10 units
Total value of output	₹5	₹10

Additional capital investment: ₹150 − ₹100 = ₹50.
Additional output: ₹10 − ₹5 = ₹5.

Calculation: ICOR = 50 ÷ 5 = 10.

Interpretation: ₹10 of additional capital investment is required to generate ₹1 of additional output in this example.

COR versus ICOR:

Basis	COR	ICOR
Measures	Total capital relative to total output	Additional capital relative to additional output
Formula	Total capital ÷ Total output	Additional capital ÷ Additional output
Example result	100 ÷ 5 = 20	50 ÷ 5 = 10

Assumption: Additional investment and additional output should be measured over comparable periods. The simplified example assumes productive investment and the stated increase in output.

8. ICOR and Capital Efficiency

ICOR is commonly used as an indicator of how efficiently additional capital is converted into additional output.

Lower ICOR: Less additional capital is required to generate one unit of additional output. This generally indicates higher capital efficiency.

Higher ICOR: More additional capital is required to generate one unit of additional output. This generally indicates lower capital efficiency.

Country comparison:

Particular	Country A	Country B
Additional investment	₹1,000 crore	₹1,000 crore
Additional GDP	₹500 crore	₹200 crore
ICOR	1,000 ÷ 500 = 2	1,000 ÷ 200 = 5

Conclusion: Country A has a lower ICOR of 2 than Country B's ICOR of 5. Based on this indicator, Country A converts additional investment into additional output more efficiently.

Exam point: Lower ICOR generally indicates higher capital efficiency; higher ICOR generally indicates lower capital efficiency. ICOR is not a perfect or standalone measure of efficiency.

9. ICOR as a Catch-All Expression

Core idea: ICOR is often described as a **catch-all expression** because it is influenced by several factors determining how effectively additional investment is converted into additional output. It does not measure only the quantity or quality of physical capital.

Example – same machinery, different labour skills: Two factories make the same additional investment of ₹100 crore in identical machinery. The machinery itself has no defect.

Particular	Factory A	Factory B
Additional investment	₹100 crore	₹100 crore
Labour	Skilled and trained	Insufficiently skilled
Use of machinery	Efficient	Inefficient
Additional output	₹25 crore	₹10 crore
ICOR	100 ÷ 25 = 4	100 ÷ 10 = 10

Explanation: In Factory B, workers cannot operate the machinery efficiently because of inadequate skills. The same additional investment therefore generates less additional output. Since ICOR is additional investment divided by additional output, lower output causes ICOR to rise.

Important correction: Poor labour skills do not directly increase ICOR. They reduce the additional output generated from a given amount of additional investment, which consequently increases ICOR.

Factors influencing ICOR:

1. **Governance and institutional quality:** Weak administration, delays, regulatory uncertainty and poor coordination can reduce the productive use of investment.
2. **Quality of labour:** Education, vocational training and skill development affect workers' ability to use machinery and capital effectively.
3. **Technology:** Outdated technology or inability to adopt and use modern technology can limit additional output.
4. **Infrastructure:** Inadequate roads, power, logistics, water supply and digital connectivity can prevent machinery and factories from operating efficiently.
5. **Capacity utilisation:** Underutilised factories or machinery may generate less output from available capital.
6. **Project implementation and management:** Cost overruns, delays, poor planning and inefficient management can reduce the output obtained from investment.
7. **Complementary inputs:** Shortages of raw materials, energy, skilled technicians or other essential inputs can constrain production even when capital is available.

Exam-oriented conclusion: A higher ICOR may indicate lower capital efficiency, but it does not necessarily mean that the physical capital itself is defective or poor in quality. It may reflect weaknesses in labour skills, technology, governance, infrastructure, capacity utilisation and other complementary factors. Hence, ICOR is a broad aggregate indicator, not a pure measure of capital efficiency.

Qualification: ICOR is also affected by the time lag between investment and resulting output, the composition of investment and broader economic conditions. Comparisons should therefore be made carefully.

10. Relationship Between ICOR, Investment Rate and Economic Growth

ICOR is used in economic planning to estimate the growth that may be achieved from a given investment rate, or the investment required to achieve a target growth rate. These are simplified relationships that assume a reasonably stable ICOR.

A. Estimating economic growth from the investment rate

Investment rate means investment expressed as a percentage of GDP.

$$\text{Economic Growth Rate} = \frac{\text{Investment Rate}}{\text{ICOR}}$$

Example: If the investment rate is 30% of GDP and ICOR is 5, estimated growth = 30 ÷ 5 = 6%.

B. Estimating additional investment required for a target growth rate

Assume: Existing GDP = ₹1,000 crore; target growth rate = 7%; ICOR = 4.

$$\text{Additional GDP} = \text{Existing GDP} \times \text{Targeted Growth Rate} \div 100$$

Step 1 – Additional output: ₹1,000 crore × 7 ÷ 100 = ₹70 crore.

$$\text{Additional Investment} = \text{ICOR} \times \text{Additional Output}$$

Step 2 – Additional investment: 4 × ₹70 crore = ₹280 crore.

Interpretation: Under the assumed ICOR of 4, additional investment of ₹280 crore is estimated to generate ₹70 crore of additional GDP, corresponding to the 7% target.

C. Estimating the investment rate required for a target growth rate

Investment Rate = Targeted Growth Rate × ICOR

Example: Targeted growth = 7%; ICOR = 4. Required investment rate = 7 × 4 = 28% of GDP.

Summary of relationships:

Purpose	Relationship
Calculate ICOR	Additional capital investment ÷ Additional output
Estimate economic growth	Investment rate ÷ ICOR
Required investment rate	Targeted growth rate × ICOR
Additional investment	ICOR × Additional output

Important limitation: These relationships are simplified planning tools. Actual growth also depends on labour, technology, governance, infrastructure, capacity utilisation, project delays and other factors. The assumed ICOR may change over time.

Quick Revision – Key Takeaways

• **Nominal GDP:** measured at current-year prices; affected by price and quantity changes.
• **Real GDP:** measured at base-year prices; removes the effect of price changes.
• **Base year:** reference year whose prices are used for constant-price calculations.
• **Current-price growth:** nominal growth; includes price effects.
• **Constant-price growth:** real growth; removes price effects.
• **COR:** total capital in relation to total output.
• **ICOR:** additional capital in relation to additional output.
• **Lower ICOR:** generally indicates greater capital efficiency; higher ICOR generally indicates lower capital efficiency.
• **ICOR is a catch-all expression:** influenced by labour skills, education, technology, governance, infrastructure, capacity utilisation and other factors.
• **Planning relationship:** Growth rate ≈ Investment rate ÷ ICOR, assuming ICOR is stable.

— End of Notes —

NATIONAL INCOME ACCOUNTING

ADVANCED MCQ TEST

SARVAN SIR ECONOMY NOTES

UPSC • APPSC GROUP 1 • TGPSC GROUP 1

Questions: 15	Suggested time: 25 minutes	Maximum marks: 30

INSTRUCTIONS

1. Each question has one correct answer. Select the most appropriate option.
2. Each question carries 2 marks. Maximum score: 30 marks.
3. Suggested time: 25 minutes. The answer key and explanations are placed after all questions.

SECTION I — QUESTIONS

Q1. With reference to the concepts of Nominal GDP and Real GDP, consider the following statements:

1. Nominal GDP can rise even when the physical volume of production remains unchanged.
2. Real GDP is calculated by valuing current-year quantities at base-year prices.
3. An increase in Nominal GDP necessarily indicates an increase in real output.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Q2. A country produces 100 units of a good in both 2025 and 2026. Its price rises from ₹10 to ₹12 per unit. If 2025 is the base year, which one of the following is correct?

(a) Nominal GDP rises by 20%, while Real GDP remains unchanged.
(b) Nominal GDP remains unchanged, while Real GDP rises by 20%.
(c) Both Nominal GDP and Real GDP rise by 20%.
(d) Both Nominal GDP and Real GDP remain unchanged.

Q3. With reference to the selection of an ideal base year, consider the following statements:

1. It should represent relatively normal economic conditions.
2. Prices of all goods and services must remain unchanged throughout the base year.
3. Reliable and comprehensive statistical data should be available for the year.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) 1, 2 and 3

Q4. Which one of the following best explains the purpose of periodically revising the base year in National Accounts?

(a) To ensure that measured GDP rises every time the base year is revised.
(b) To remove all fluctuations in market prices from the economy.
(c) To incorporate changes in economic structure, updated data sources and improved statistical methods.
(d) To make Nominal GDP and Real GDP numerically identical.

Q5. National income at current prices rises from ₹1,000 crore to ₹1,200 crore, while national income at constant prices rises from ₹1,000 crore to ₹1,050 crore. Which inference is the most appropriate?

(a) Nominal growth is 5% and real growth is 20%.
(b) Nominal growth is 20% and real growth is 5%; the current-price measure includes price effects.
(c) Both nominal and real growth are 20%.
(d) Real national income has declined by 15%.

Q6. According to the traditional Indian national income concept, consider the following statements:

1. National income is measured as Net National Product at Factor Cost (NNP at FC).
2. GDP and national income are identical concepts.
3. GDP growth and national income growth should not be used interchangeably without specifying the measure.

Which of the statements given above are correct?

(a) 1 only
(b) 1 and 2 only
(c) 1 and 3 only
(d) 2 and 3 only

Q7. A production unit employs total capital of ₹240 crore and generates output worth ₹12 crore. What is its Capital-Output Ratio (COR)?

(a) 0.05
(b) 5
(c) 20
(d) 28

Q8. A factory's total capital increases from ₹100 crore to ₹160 crore. Over a comparable period, the total value of output increases from ₹5 crore to ₹15 crore. What is its ICOR?

(a) 4
(b) 5
(c) 6
(d) 12

Q9. Consider the following statements distinguishing COR from ICOR:

1. COR relates total capital employed to total output.
2. ICOR relates additional capital investment to additional output.
3. A lower ICOR must always be accompanied by a lower COR.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Q10. Two countries each make additional investment of ₹1,000 crore. Country P generates additional GDP of ₹500 crore, while Country Q generates additional GDP of ₹200 crore. Which statement is correct?

(a) P has ICOR 5 and Q has ICOR 2; Q is more capital-efficient.
(b) P has ICOR 2 and Q has ICOR 5; P shows higher capital efficiency on this indicator.
(c) Both have ICOR 2.5 and equal capital efficiency.
(d) P has ICOR 0.5 and Q has ICOR 0.2; Q is less capital-efficient.

Q11. Two factories invest the same additional amount in identical, functioning machinery. Factory A has trained workers and generates additional output of ₹25 crore; Factory B has insufficiently skilled workers and generates ₹10 crore. Which inference best follows?

(a) Factory B's ICOR is lower because its workers require less training.
(b) Factory B's ICOR is higher because the same additional investment generates less additional output.
(c) The ICORs must be equal because the machinery is identical.
(d) Labour skills cannot affect ICOR if the physical capital is identical.

Q12. ICOR is sometimes described as a 'catch-all expression'. Consider the following factors that can influence it:

1. Governance and institutional quality
2. Education, vocational training and labour skills
3. Technology and infrastructure
4. Capacity utilisation and project implementation

Which of the factors listed above can influence ICOR?

(a) 1 and 2 only
(b) 1, 2 and 3 only
(c) 2, 3 and 4 only
(d) 1, 2, 3 and 4

Q13. Which one of the following statements about a rise in ICOR is the most precise?

(a) It proves that the quality of physical machinery has deteriorated.
(b) It necessarily means that the amount of investment has fallen.
(c) It may reflect lower additional output due to labour, technology, governance, infrastructure or other constraints, even if machinery itself is not defective.
(d) It proves that the economy's real GDP has contracted.

Q14. If the investment rate is 32% of GDP and ICOR is 4, the simplified planning relationship implies an estimated economic growth rate of:

(a) 4%
(b) 8%
(c) 12%
(d) 128%

Q15. An economy has existing GDP of ₹2,000 crore and targets growth of 6%. If ICOR is assumed to be 3, which pair correctly gives the additional GDP and the estimated additional investment required?

(a) ₹60 crore and ₹180 crore
(b) ₹120 crore and ₹360 crore
(c) ₹120 crore and ₹40 crore
(d) ₹600 crore and ₹1,800 crore

SECTION II — ANSWER KEY & DETAILED EXPLANATIONS

ANSWER KEY

1. A	2. A	3. B	4. C	5. B
6. C	7. C	8. C	9. A	10. B
11. B	12. D	13. C	14. B	15. B

DETAILED EXPLANATIONS

Q1. Correct answer: (a)

Statements 1 and 2 are correct. Nominal GDP reflects both price and quantity changes, so it can rise solely because prices increase. Real GDP values current-year quantities at base-year prices. Statement 3 is incorrect: nominal GDP may rise without any increase in real output.

Q2. Correct answer: (a)

Nominal GDP rises from ₹1,000 (100 × ₹10) to ₹1,200 (100 × ₹12), an increase of 20%. Real GDP remains ₹1,000 in both years because the base-year price of ₹10 is used and the quantity remains 100 units.

Q3. Correct answer: (b)

Statements 1 and 3 are correct. An ideal base year should reflect normal conditions and have reliable, comprehensive data. Statement 2 is incorrect: prices need not remain unchanged during the base year; the year should provide a representative benchmark.

Q4. Correct answer: (c)

Base-year revision incorporates changes in the structure of the economy, updated data sources and improved estimation methods. It is intended to improve the relevance and accuracy of national accounts, not to guarantee a higher GDP figure or make nominal and real GDP identical.

Q5. Correct answer: (b)

Current-price growth = [(1,200 − 1,000) / 1,000] × 100 = 20%. Constant-price growth = [(1,050 − 1,000) / 1,000] × 100 = 5%. Current-price growth reflects price and output changes, while constant-price growth removes the price-change effect.

Q6. Correct answer: (c)

Statements 1 and 3 are correct in the traditional terminology used in the notes. National income is expressed as NNP at Factor Cost. GDP and national income are not identical concepts; therefore, their growth rates should not be treated as interchangeable.

Q7. Correct answer: (c)

COR = Total Capital Employed / Total Value of Output = 240 / 12 = 20. Thus, ₹20 of capital is employed to generate ₹1 worth of output.

DETAILED EXPLANATIONS (CONTINUED)

Q8. Correct answer: (c)

Additional capital = ₹160 − ₹100 = ₹60 crore. Additional output = ₹15 − ₹5 = ₹10 crore. ICOR = Additional Capital / Additional Output = 60 / 10 = 6.

Q9. Correct answer: (a)

Statements 1 and 2 correctly distinguish the ratios. Statement 3 is incorrect. COR uses total capital and total output, whereas ICOR uses changes in capital and output. The definitions do not require the two ratios to move together.

Q10. Correct answer: (b)

Country P: ICOR = 1,000 / 500 = 2. Country Q: ICOR = 1,000 / 200 = 5. A lower ICOR generally indicates that less additional capital is needed per unit of additional output. Hence, P shows higher capital efficiency on this indicator.

Q11. Correct answer: (b)

For the same additional investment, lower additional output produces a higher ICOR. If each factory invests ₹100 crore, A's ICOR is 100 / 25 = 4, while B's is 100 / 10 = 10. Identical machinery does not guarantee identical output because labour skills affect how effectively capital is used.

Q12. Correct answer: (d)

All four can influence the conversion of additional investment into additional output. Governance, labour skills, technology, infrastructure, capacity utilisation and project management affect the productivity of investment. Shortages of complementary inputs can also constrain output.

Q13. Correct answer: (c)

A higher ICOR may indicate lower capital efficiency, but it does not prove that machinery is defective. Weak labour skills, technology, governance, infrastructure, capacity utilisation or complementary inputs may reduce the additional output generated by a given investment.

Q14. Correct answer: (b)

Estimated growth rate = Investment Rate / ICOR = 32 / 4 = 8%. This is a simplified planning relationship and assumes a reasonably stable ICOR.

Q15. Correct answer: (b)

Additional GDP = Existing GDP × Target Growth Rate / 100 = 2,000 × 6 / 100 = ₹120 crore. Additional investment = ICOR × Additional Output = 3 × 120 = ₹360 crore.

SARVAN SIR ECONOMY NOTES

NATIONAL INCOME & MACROECONOMIC CONCEPTS

1. GDP DEFLATOR

Definition

GDP Deflator is a price index that measures the overall change in the prices of all final goods and services produced within the domestic territory of an economy, compared to the prices prevailing in the base year.

In simple terms, it measures the change in the general price level of domestically produced final goods and services.

Formula

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

- Nominal GDP: Value of current-year production calculated at current-year prices.
- Real GDP: Value of current-year production calculated at base-year prices.

Illustrative Example

Suppose an economy produces the following quantity of a particular good:

Particulars	Base Year	Current Year
Quantity produced	100 units	120 units
Price per unit	Rs. 10	Rs. 12

Step 1: Calculation of Nominal GDP

Nominal GDP = Current-year quantity × Current-year price

= 120 × Rs. 12 = Rs. 1,440

Step 2: Calculation of Real GDP

Real GDP = Current-year quantity × Base-year price

= 120 × Rs. 10 = Rs. 1,200

Step 3: Calculation of GDP Deflator

$$\text{GDP Deflator} = \frac{\text{Rs. 1,440}}{\text{Rs. 1,200}} \times 100 = 120$$

Interpretation: A GDP Deflator of 120 indicates that the overall price level is 20% higher than that of the base year.

Important Points

- GDP Deflator measures changes in the overall price level of domestically produced final goods and services.
- It includes both consumer goods and capital goods.
- Imported goods are excluded because they are not produced within the domestic territory.
- Unlike CPI, GDP Deflator is not based on a fixed basket of goods and services.
- In the base year, GDP Deflator is 100 because Nominal GDP and Real GDP are equal.

2. POTENTIAL GDP

Definition

Potential GDP refers to the maximum level of real output that an economy can sustainably produce when its available productive resources are employed at their normal and sustainable levels, without generating additional inflationary pressure.

In simple terms, Potential GDP represents the sustainable productive capacity of an economy.

Illustrative Example

Suppose the Potential GDP of an economy is Rs. 200 lakh crore, whereas its Actual GDP is Rs. 180 lakh crore.

- Potential GDP: Rs. 200 lakh crore
- Actual GDP: Rs. 180 lakh crore

Interpretation: The economy is producing below its potential productive capacity, suggesting underutilisation of resources.

Factors Determining Potential GDP

- Availability of labour.
- Availability of physical capital.
- Technological advancement.
- Labour productivity and skill development.
- Human capital and education.
- Infrastructure development.

Improvements in these factors can increase the productive capacity of an economy and thereby raise Potential GDP.

Actual GDP vs Potential GDP

Actual GDP	Potential GDP
Represents the actual output produced by an economy.	Represents the sustainable productive capacity of an economy.
Can fluctuate depending on aggregate demand and supply conditions.	Depends mainly on productive resources and productivity.
Can be below, equal to or above Potential GDP.	Represents the economy's sustainable output level.

3. GREEN GDP

Definition

Green GDP is an adjusted measure of GDP that accounts for environmental degradation and the depletion of natural resources while calculating the economic output of a country.

In simple terms, Green GDP is calculated by deducting the economic value of environmental damage and natural resource depletion from conventional GDP.

Formula

Green GDP = Conventional GDP
− Cost of Environmental Degradation
− Cost of Natural Resource Depletion

Illustrative Example

Suppose an economy has the following values:

- Conventional GDP = Rs. 200 lakh crore

- Cost of environmental degradation = Rs. 15 lakh crore
- Cost of natural resource depletion = Rs. 5 lakh crore

Calculation of Green GDP

Green GDP = 200 − 15 − 5

Green GDP = Rs. 180 lakh crore

Interpretation: After deducting the estimated costs of environmental degradation and natural resource depletion, the Green GDP of the economy is Rs. 180 lakh crore.

Important Points for Prelims

- Green GDP accounts for environmental costs while measuring economic performance.
- It considers environmental degradation, such as air and water pollution.
- It accounts for the depletion of natural resources, such as forests and minerals.
- Green GDP is generally lower than conventional GDP when environmental costs and resource depletion are positive.
- It provides a better indication of environmentally sustainable economic growth.

Key distinction: Conventional GDP measures economic production, whereas Green GDP adjusts that measure to account for environmental costs.

SARVAN SIR ECONOMY NOTES

IMPORTANT FACTS RELATED TO NATIONAL INCOME ACCOUNTING

Competitive Examination Material | UPSC • APPSC • TGPSC

1. Historical Background of National Income Estimation in India

First Attempt – Dadabhai Naoroji

- The first attempt to calculate the National Income of India was made in 1867–1868 by Dadabhai Naoroji.
- In his book Poverty and Un-British Rule in India, he estimated India's per capita annual income at ₹20.

First Scientific Estimate – Professor V. K. R. V. Rao

- The first scientific estimate of National Income was made by Professor V. K. R. V. Rao in 1931–32.

2. National Income Committee (1949)

- Soon after Independence, the Government of India appointed the National Income Committee in August 1949.
- The Committee was constituted under the chairmanship of Professor P. C. Mahalanobis.
- The other two members were Professor D. R. Gadgil and Dr. V. K. R. V. Rao.
- The Committee was set up to compile estimates of National Income.
- It submitted its final report in 1954, giving estimates of National Income for the period 1950–1954.
- According to the report, total National Income was estimated at ₹8,650 crore, and per capita income at ₹246.90.

Definition of National Income – National Income Committee (1949)

Definition: “A National Income estimate measures the volume of commodities and services turned out during a given period, counted without duplication.”

3. Institutions Involved in National Income Estimation

Central Statistical Organisation (CSO)

- The Central Statistical Organisation (CSO) was set up on 2 May 1951.
- For further estimation of National Income, the Government established the CSO.

National Sample Survey Office (NSSO)

- The National Sample Survey Office (NSSO) was set up in 1950.
- It was established to conduct large-scale sample surveys to meet the country's data requirements for estimating National Income and other aggregates.

National Statistical Commission (NSC)

- The National Statistical Commission was set up by the Government of India through a resolution dated 1 June 2005, on the recommendations of the Rangarajan Commission.

4. Changes in the Base Year of National Accounts

- In 2015, the Central Statistical Office introduced a new series of National Accounts Statistics.
- The base year of the GDP series was revised from 2004–05 to 2011–12.

5. Merger of CSO and NSSO

- The Government of India merged the Central Statistical Office (CSO) and the National Sample Survey Office (NSSO) into a single entity on 23 May 2019.
- The newly merged entity was named the National Statistical Office (NSO).

6. Ministry of Statistics and Programme Implementation (MoSPI)

- The Ministry of Statistics and Programme Implementation came into existence as an independent ministry on 15 October 1999, after the merger of the Department of Statistics and the Department of Programme Implementation.
- At present, the National Statistical Office (NSO), under MoSPI, is responsible for the estimation and publication of National Income.
- The new series incorporates the latest recommendations of the System of National Accounts (SNA), 2008, the international guidelines on the compilation of National Accounts.

7. GDP Calculation Methodology by NSO

The NSO calculates GDP using both the Value Added Method and the Expenditure Method.

A. Value Added Method

Under the Value Added Method, GDP is calculated by measuring the value addition made by various economic activities.

The economic activities listed in the source material are:

- Agriculture, forestry and fishing
- Mining and quarrying
- Manufacturing
- Electricity, gas, water supply and other utility services
- Construction
- Trade, hotels and transport
- Communication and services related to broadcasting
- Financial, insurance, real estate and professional services
- Public administration, defence and other services

B. Expenditure Method

Under the Expenditure Method, GDP is calculated by adding the expenditure incurred by the different sectors of the economy.

The expenditure components are:

Expenditure Component	Share shown
Private Final Consumption Expenditure (PFCE)	60%
Government Expenditure	12%
Gross Fixed Capital Formation (GFCF)	32%
Net Exports	−4%
Total	100%

C. Discrepancies between the Two Methods

The textbook notes that GDP measured from either side should theoretically be equal. However, because reliable data may not be available for every component, a difference can arise between the two methods. This difference is referred to as discrepancies.

D. GDP / GSDP Aggregation Note

India’s GDP includes the output of States as well as the output of specified Central Government activities and Indian residents operating in international waters:

> GDP of India = All States’ Gross State Domestic Product (GSDP) + output from Centre-specific activities (such as Railways, Defence and Central Highways) + output from activities such as fishing vessels, oil rigs and floating platforms operated by residents of India in international waters.

MODEL MAINS QUESTIONS

GDP Calculation Methods • Investment • Circular Flow of Income

Exam orientation: APPSC Group 1 Mains | TGPSC Group 1 Mains | UPSC GS

Note: These are model questions framed with reference to the previous-year question pattern. They are not verbatim previous-year questions.

1. VALUE-ADDED METHOD

10 marks | 150 words

"The value-added method of calculating GDP helps avoid the problem of double counting in an economy." Explain the statement. Discuss the significance of intermediate goods and final goods in measuring national income.

> **Answer pointers:** Value added; intermediate goods; final goods; double counting; sectoral contribution; illustrative example.

2. INCOME METHOD

10 marks | 150 words

Explain how national income is estimated through the income method. Discuss the major components of factor income and the difficulties involved in estimating income generated in an economy like India.

> **Answer pointers:** Compensation of employees; operating surplus; mixed income; self-employed persons; informal sector; treatment of depreciation and indirect taxes.

3. EXPENDITURE METHOD

10 marks | 150 words

"The expenditure method measures GDP by aggregating expenditure on final goods and services in an economy." Explain its components and examine the significance of net exports in calculating GDP.

> **Answer pointers:** Private final consumption expenditure; government final consumption expenditure; gross capital formation; exports; imports; GDP expenditure identity.

4. INVESTMENT

10 marks | 150 words

"Investment in an economy is fundamentally associated with capital formation rather than merely financial transactions." Explain the statement. Differentiate between gross investment, net investment and capital stock.

> **Answer pointers:** Capital goods; GFCF; GCF; depreciation; net investment; why the purchase of existing shares does not constitute current productive investment.

5. CIRCULAR FLOW OF INCOME

10 marks | 150 words

Explain the circular flow of income in a two-sector economy. How do savings and investment operate as leakages and injections, respectively, and what is their significance in maintaining equilibrium in the economy?

> **Answer pointers:** Households; firms; factor services; factor incomes; consumption expenditure; real and monetary flows; savings; investment; ex-post S = I in the simplified model.

SUGGESTED PRACTICE

Write each answer in approximately 150 words. Allow 8–10 minutes per question. Focus on conceptual clarity, correct economic terminology, a logical structure and a brief conclusion.